



National University of Computer and Emerging Sciences, Lahore



FictionGenie

Muhammad Usman Azeem 20L-2064 BS(CS)

Muhammad Usama 20L-1233 BS(CS)

Zaid Khan 20L-2072 BS(CS)

Supervisor: Mr. Waqas Manzoor

Final Year Project

June 9, 2024

Anti-Plagiarism Declaration

This is to declare that the above publication was produced under the:

Title: FictionGenie

is the sole contribution of the author(s), and no part hereof has been reproduced as it is the basis (cut and paste) that can be considered Plagiarism. All referenced parts have been used to argue the idea and cited properly. I/We will be responsible and liable for any consequence if a violation of this declaration is determined.

Date: .....

Name: Muhammad Usama

Signature: .....

Name: Muhammad Usman Azeem

Signature: .....

Name: Zaid Khan

Signature: .....

Author's Declaration

This states Authors' declaration that the work presented in the report is their own, and has not been submitted/presented previously to any other institution or organization.

Abstract

The classical art of storytelling is undergoing a remarkable transformation in this evolving digital era, thanks to the emergence of AI. FictionGenie stands at the forefront of this revolution, offering an innovative AI-powered storytelling platform tailored specifically for children. This mobile application will foster cognitive-linguistic development in children. Our application is designed with a user-centric approach, an attractive interface, and dynamic themes. FictionGenie offers robust parental monitoring which includes content curation, setting usage limits, and viewing children's learning insights. To embrace diversity, we have incorporated a multilingual feature to read stories. The architectural blueprint prioritizes scalability and the seamless integration of new features and content. FictionGenie is poised to shape the future of storytelling in this digital era.

Executive Summary

In the era of Artificial Intelligence storytelling is undergoing a profound transformation. FictionGenie is poised to revolutionize storytelling for children. Our platform stands as a testament to the transformative power of AI in education and entertainment. FictionGenie, our mobile application, is specifically designed to not only entertain children but improve their cognitive and linguistic skills. The sole purpose of FictionGenie is to create an interactive application enabling children to read stories of their own choices and recommendations.

FictionGenie is equipped with a lot of features for its users to bring a revolution in the field of AI-generated stories. A wide range of interesting stories from different genres and languages are offered. Our multilingual support feature allows kids to experience the magic of storytelling in their own languages making our application useful for a diverse audience.FictionGenie is a platform that uses AI algorithms to customize reading recommendations for children, regardless of their reading ability. It offers text-to-speech technology and strong parental controls, ensuring a secure online experience. The platform exposes kids to various storylines, stimulating their imaginations and improving their reading skills. FictionGenie creates engaging visuals that match the narrative, enhancing children's engagement and creativity. The platform's user-centric approach and visually appealing interface ensure a seamless reading experience. With no ads and a free, user-friendly interface, FictionGenie is a new and innovative way to engage young readers. There are some notable challenges associated with the development of our project as generating content with the use of AI requires a lot of evaluation and research so for this reason, we have researched a lot of literature and reports out of which some major ones are highlighted in chapter 3. According to our research, the main challenge would not only be that the content our application generated is according to the prompts of the user but we have to ensure that the user is completely engaged with the content as the stories should be unique and are narrated in a way that these are easy to understand and engaging to users. The sentence structure of the stories should be tailored in a way that gives the experience to the user that the content of the story is written by a human, making the reading experience both relatable and enriching.

The software requirement specification of FictionGenie prioritize content filtering for parental control and for children, different APIs to provide multiple feature to our young users. These specifications include multilingual story reading, instant word definition/meaning, audio story books and Gemini to generate stories instantly. Fictions will also provide dynamic user experience by sorting and filtering genres based on user preferences. Our purposed methadology includes agile development that will ensure adaptability and scalability with planned release on App Store and Play Store. Our responsive and user-centric design will contribute to unwavering commitment to user satisfaction with the aim to capti-

vate young minds globally. The highlevel and lowlevel system design of FictionGenie inventively links the key components. These components include user-authentication for parents and childeren, APIs to instantly generate stories, translate them to multiple languages, dynamic image creation based on genre of the tale, sentence dictionaries and allow our young users to listen to these tales. FictionGenie will be built on versatile and scalable Flutter framework which will allow user-centric design principles with charming user interface, animations and apealing components to create an immersive and pleasent user experience.

In the future, the research aims to revolutionize children's storytelling by increasing user engagement through interactive controls and emotional intelligence integration. Natural language generation techniques will be improved in order to produce narratives that are similar to those of people, and they will be modified in accordance with reader preferences to provide a unique and engaging reading experience. The importance of ethical standards and cultural diversity will be stressed when dealing with various issues and languages. Children will be empowered by collaborative storytelling models, which will also develop creativity and correlate with educational objectives. Making FictionGenie an ever-evolving, culturally aware, and educationally stimulating AI-driven platform for young readers, FictionGenie will take continual feedback integration and machine learning analysis.

Table of Contents

List of Figures **xii**

List of Tables **xiii**

1 Introduction **1**

1.1 Purpose of this Document 1

1.2 Intended Audience 1

1.3 Definitions and Acronyms 2

1.3.1 Definitions 2

1.3.2 Acronyms 2

1.4 Conclusion 2

2 Project Vision **3**

2.1 Problem Domain Overview 3

2.2 Problem Statement 3

2.3 Problem Elaboration 3

2.4 Goals and Objectives 4

2.5 Project Scope 4

2.6 Sustainable Development Goal (SDG) 5

2.6.1 SDG-4 (Quality Education) 5

2.6.2 SDG-9 (Industry, Innovation, and Infrastructure) 5

2.7 Business Opportunity 6

2.8 Conclusion 6

3 Literature Review / Related Work **7**

3.1 Definitions and Acronyms 7

3.1.1 Definitions 7

3.1.2 Acronyms 8

3.2 Detailed Literature Review 8

3.2.1 Evaluating Story Generation Systems Using Automated Linguistic Analyses . . 8
3.2.2 Interactive Storytelling: From AI Experiment To New Media 10
3.2.3 Collaborative Storytelling with Human Actors and AI Narrators 12
3.2.4 Story Generation from Sequence of Independent Short Descriptions 13
3.2.5 Automatic Story Generation: Challenges and Attempts 15
3.2.6 Learning to Tell Tales: A Data-Driven Approach to Story Generation 16
3.2.7 Development of an API to Create Interactive Storytelling Systems 17
3.2.8 Design implications of generative AI systems for visual storytelling for young learners 19
3.2.9 Ambient Adventures: Teaching ChatGPT on Developing Complex Stories 20
3.2.10 Controllable Text to Image Generation with GPT-4 21
3.3 Literature Review Summary Table 22
3.4 Related Work 24
3.4.1 AI Story Generator and Maker 24
3.4.2 Story Spark 25
3.4.3 Kidstory: AI bedtime stories 26
3.4.4 Story Seed 27
3.5 Related Work Summary Table 28
3.6 Conclusion 29
3.7 Selected Model 29

4 Software Requirement Specifications 30
4.1 List of Features 30
4.2 Functional Requirements 30
4.2.1 Functional Requirements of Children 30
4.2.2 Functional Requirements of Parents 30
4.2.3 Functional Requirements of System 31
4.3 Quality Attributes 31
4.3.1 Reliability 31
4.3.2 Maintainability 31
4.3.3 Usability 31
4.3.4 Correctness 31
4.3.5 Efficiency 32
4.3.6 Flexibility 32
4.4 Non-Functional Requirements 32

4.4.1 Performance . . . 32
4.4.2 Reliability . . . 32
4.4.3 Security . . . 32
4.4.4 Compatibility . . . 33
4.5 Assumptions . . . 33
4.6 Use Cases . . . 33
4.6.1 User Registration . . . 33
4.6.2 User Login . . . 34
4.6.3 Story Generation . . . 34
4.6.4 Parental Control . . . 35
4.6.5 Multilingual Story Selection . . . 35
4.6.6 Text-to-Speech . . . 36
4.6.7 Dynamic Image Generation . . . 36
4.6.8 App Interface Dark/Light Mode . . . 37
4.6.9 Story Recommendation . . . 37
4.6.10 Instant Word Definition . . . 38
4.6.11 Add New Story in the Application . . . 38
4.6.12 Read Pre-built Stories . . . 39
4.6.13 Update Profile . . . 39
4.7 Hardware and Software Requirements . . . 40
4.7.1 Hardware Requirements . . . 40
4.7.2 Software Requirements . . . 40
4.8 Graphical User Interface . . . 40
4.8.1 Welcome Page . . . 41
4.8.2 Login Page . . . 41
4.8.3 Signup Page . . . 42
4.8.4 Forgot Password Page . . . 42
4.8.5 Profile Avatar Selection Page . . . 43
4.8.6 Home Page . . . 43
4.8.7 Story Generation Page . . . 44
4.8.8 Story Page . . . 44
4.8.9 Story Page in Dark Mode . . . 45
4.8.10 Explore Story Page . . . 45
4.8.11 Recently Viewed Stories Page . . . 46

4.8.12 Favourite Stories Page . . . 46
4.9 Database Design . . . 47
4.9.1 ER Diagram . . . 47
4.9.2 Data Dictionary . . . 48
4.10 Risk Analysis . . . 49
4.11 Conclusion . . . 49

5 Proposed Approach and Methodology 50
5.1 Story Generation with Google Gemini API . . . 50
5.2 Agile Development . . . 50
5.3 Text to Speech API . . . 50
5.4 Dynamic Image Generation . . . 50
5.5 Multilingual Support . . . 51
5.6 User Centric Design . . . 51
5.7 Conclusion . . . 51

6 High-Level and Low-Level Design 52
6.1 System Overview . . . 52
6.1.1 AI-Based Story Generation . . . 52
6.1.2 AI-Generated Images . . . 52
6.1.3 Pre-built Stories Library . . . 52
6.1.4 Story Narration with Text-to-Speech . . . 53
6.1.5 Multilingual Support . . . 53
6.1.6 Age-Specific Content . . . 53
6.1.7 Parental Monitoring . . . 53
6.2 Design Considerations . . . 53
6.2.1 Assumption and Dependencies . . . 53
6.2.2 General Constraints . . . 54
6.2.3 Goals and Guidelines . . . 58
6.2.4 Development Methods . . . 58
6.3 System Architecture . . . 59
6.3.1 User Interface . . . 60
6.3.2 Server . . . 60
6.3.3 User Management and Authentication . . . 60
6.3.4 Story Tailoring . . . 60

6.3.5 Content Fetching and Storage . . . 60
6.3.6 Parental Control and Monitoring . . . 60
6.3.7 Python Server . . . 60
6.3.8 Recommendation Engine . . . 60
6.3.9 Third-Party Services . . . 61
6.4 Architectural Strategies . . . 61
6.4.1 Programming Languages . . . 61
6.4.2 Cloud-Based Database . . . 61
6.4.3 Error detection and Recovery . . . 61
6.4.4 AI Integration with Large Language Models (Gemini) . . . 61
6.4.5 Scalability and Load Balancing . . . 62
6.4.6 Concurrency and Load Balancing . . . 62
6.5 Domain Model/Class Diagram . . . 62
6.6 Sequence Diagram . . . 63
6.7 Policies and Tactics . . . 69
6.7.1 Tools to be Used . . . 69
6.7.2 Coding guidelines and conventions . . . 70
6.7.3 User Interface . . . 70
6.7.4 Extensibility . . . 70
6.7.5 Policies for System Testing . . . 70
6.7.6 Software maintenance . . . 70
6.7.7 Algorithms . . . 70
6.8 Conclusion . . . 71

7 Implementation and Test Cases 72

7.1 Implementation . . . 72
7.1.1 Pre-built Stories Library . . . 72
7.1.2 Exploring AWS Lambda . . . 72
7.1.3 AI Based Stories . . . 72
7.1.4 AI Generated Images . . . 73
7.1.5 User Registration . . . 73
7.1.6 User Authentication . . . 73
7.1.7 Profile Customization . . . 73
7.1.8 Favorite Stories . . . 73
7.2 Test Case Design and Description . . . 74

7.3 Test Metrics . . . 81
7.4 Conclusion . . . 81

8 User Manual 82
8.1 Getting Started . . . 82
8.2 Login . . . 82
8.3 Signup . . . 82
8.4 Forgot Password . . . 83
8.5 Reading a Pre-Built Story . . . 83
8.6 Generating an AI-based Story . . . 84
8.7 Add to Favourites . . . 84
8.8 Parental Monitoring . . . 85
8.9 Edit Account Details . . . 85
8.10 Account Deletion . . . 86
8.11 Contacting the FictionGenie Team . . . 86
8.12 Conclusion . . . 86

9 Experimental Results and Discussion 88
9.1 Pre-built Stories Library . . . 88
9.1.1 User Satisfaction Survey . . . 88
9.1.2 Popular Genres . . . 88
9.1.3 Feature Popularity . . . 88
9.1.4 Usage Frequency Analysis . . . 88
9.1.5 System Result . . . 89
9.2 AI Generated Stories . . . 89
9.2.1 Quality of Stories . . . 89
9.2.2 Popular Genres . . . 89
9.2.3 User Satisfaction . . . 90
9.3 AI Generated Images . . . 90
9.3.1 Image Relevance . . . 90
9.3.2 User Satisfaction . . . 90
9.4 Conclusion . . . 91

10 Conclusion and Future Work 92

List of Figures

2.1 **Quality Education** . . . 5
2.2 **Industry Innovation and Infrastructure** . . . 5

3.1 **HTN Representation of lead character behaviour** . . . 11
3.2 **Tree Format** . . . 17
3.3 **AI Story Generator and Maker** . . . 24
3.4 **Story Spark** . . . 25
3.5 **Kidstory: AI bedtime stories** . . . 26
3.6 **Story Seed** . . . 27

4.1 **Welcome Page** . . . 41
4.2 **Login Page** . . . 41
4.3 **Signup Page** . . . 42
4.4 **Forgot Password Page** . . . 42
4.5 **Profile Avatar Selection Page** . . . 43
4.6 **Home Page** . . . 43
4.7 **Story Generation Page** . . . 44
4.8 **Story Page** . . . 44
4.9 **Story Page in Dark Mode** . . . 45
4.10 **Explore Story Page** . . . 45
4.11 **Recently Viewed Stories Page** . . . 46
4.12 **Favourite Stories Page** . . . 46
4.13 **ER Diagram** . . . 47

6.1 **System Architecture Diagram** . . . 59
6.2 **Class Diagram** . . . 62
6.3 **User Registration** . . . 63
6.4 **User Login** . . . 63
6.5 **Story Generation** . . . 64

6.6 **Parental Control** 64
6.7 **Multilingual Story Selection** 65
6.8 **Text to Speech** 65
6.9 **Dynamic Image Generation** 66
6.10 **App Interface Dark/Light Mode** 66
6.11 **Story Recommendation** 67
6.12 **Instant Word Definition** 67
6.13 **Add New Stories in the Application** 68
6.14 **Read Pre-built Stories** 68
6.15 **Update Profile** 69

List of Tables

3.1 **Summary of Literature Review** 23
3.2 **Summary of Related Work** 28

4.1 **Data Dictionary Table** 48

7.1 **Setting-Up Child Profile** 74
7.2 **Restrict Contents for Children** 75
7.3 **Child Activity Dashboard for Parents** 75
7.4 **Forget Password** 76
7.5 **Select Avatar** 76
7.6 **Read Stories** 77
7.7 **Listen to a Story** 77
7.8 **Translate a Story** 78
7.9 **Generate Story** 78
7.10 **Story Recommendation** 79
7.11 **Image Generation** 79
7.12 **Word Definition** 80
7.13 **Dark/Light Mode** 80
7.14 **Functional Test Case Metrics** 81
7.15 **Non-Functional Test Case Metrics** 81

9.1 **System Results** 89
9.2 **System Results** 90
9.3 **System Results** 91

Chapter 1 Introduction

Stories have served as a pillar of human culture and communication, passing down wisdom and knowledge from one generation to another. They hold a significant power to captivate, ignite, and inspire one's imagination. It also helps children develop a strong bond with the world of books and opens doors to new experiences and imaginative landscapes. In today's rapidly evolving digital environment, the need for personalized and age-appropriate content for children's reading experience has become increasingly important. FictionGenie is an AI-powered application designed to provide an interactive storytelling experience with dynamic themes. Our application will keep in aspect user preferences and generate stories according to their interests and hobbies which will develop curiosity amongst children to read them. The multilingual capability of the application will allow children from different cultural backgrounds to use the app easily. Our application will allow parental monitoring, enabling parents to curate content, set usage limits, and view the learning insights of their children. There are a few applications that produce children's stories upon selection of genre however, the need for this application was necessary because no such application with a combination of dynamic themes, multilingual capability, and parental monitoring was found over the internet. This application will shape the way of developing interest in books, keeping in mind technological advancements throughout the whole globe. Furthermore, rather than spending time on nonproductive applications like video games, this will help children improve their reading and learning abilities. We will also be using an integrated dictionary to allow children to learn new words and their meanings if not known to them. We aim to develop a platform for the audience that will not only entertain them but will also educate and engage with them.

1.1 Purpose of this Document

The purpose of this document is to introduce FictionGenie, an innovative application for crafting stories for children. This project will focus on developing a mobile-based application with dynamic themes and interactive user interface, shaping the way of storytelling. This document will comprehensively analyze various aspects of our project including story generation and dynamic image creation. We aim to provide a thorough insight into every aspect of our project.

1.2 Intended Audience

The intended audience for this project are children and parents. Children will be using it to read stories according to their selected preferences such as age, genre, and difficulty level of the story. Parents will have an insight into their child's usage time, the stories they have read, and new words learned by them.

1.3 Definitions and Acronyms

The Definitions and Acronyms used in this whole chapter are listed below in their sub-sections.

1.3.1 Definitions

Hierarchical Task Networks (HTN): is a planning technique in artificial intelligence where complex tasks are broken down into simpler, hierarchical subtasks. It structures tasks into a tree-like format, allowing for easier management and execution of tasks in various applications.

Phrase Based (PB): These models are based on evaluating and analyzing indvidual phrases of text, speech or data

1.3.2 Acronyms

SDG: Sustainable Development Goal

NLG: Natural Language Generation

SMT: Statistical Machine Translation

RNN: Recurrent Neural Network

AI: Artificial Intelligence

PB: Phrase Based

Seq2Seq: Sequence to Sequence

BLEU-4: Bilingual Evaluation Study

METEOR: Metric for Evaluation of Translation with Explicit Ordering

TER: Translation Edit Rate

ROUGE-L: Recall-Oriented Understudy for Gisting Evaluation

NLP: Natural Language Processing

LLM: Large Language Model

1.4 Conclusion

This report will consist of 3 chapters. The first introductory chapter defines the purpose of our project, and intended audience, and provides clear definitions of terms, acronyms, and abbreviations used throughout the whole report. The second chapter will provide an elaborative explanation of the problem statement and its domain. We will also be defining our goals and objectives.The third and last chapter of this report will focus on the literature review of the research papers selected and studied.

Chapter 2 Project Vision

FictionGenie emerges as a visionary initiative set to transform the way children encounter and engage with stories in a rapidly expanding digital ecosystem where the traditional art of storytelling intersects with the limitless potential of Artificial Intelligence (AI). Our concept envisions a future in which the old tradition of storytelling is transformed into an interactive, personalized, and educational adventure that promotes children's cognitive and language development.

2.1 Problem Domain Overview

Storytelling is experiencing a fundamental transition in the digital era, with Artificial Intelligence (AI) as the driving force. Within this arena, "FictionGenie" stands out as a trailblazing initiative, utilizing AI to create an interactive and dynamic story-telling and reading platform adapted exclusively for children. Our project's domain focuses on encouraging cognitive and linguistic growth in young brains while also delivering a dynamic and entertaining reading experience.

2.2 Problem Statement

In the past, storytelling was of great significance as it helped children improve their linguistics, creative ability, and learning skills. In today's rapidly evolving world, parents are unable to give time to their children due to their hectic schedules, therefore children lack significantly in terms of creativity. FictionGenie aims to address this issue by providing an interactive environment for children to read, learn, and improve their linguistics. To cater to personal choices, FictionGenie will be creating stories based on the preference of genre and story type selected by children to develop children's interest in stories. We will be creating an AI-powered mobile application that will educate and glorify children's knowledge.

2.3 Problem Elaboration

FictionGenie aims to reshape the storytelling experience by deploying Artificial Intelligence to create stories based on children's preferences, reading history, genre selection, and difficulty criteria selection. We also aim to break down the language barrier gap across the globe. Our application will support multilingual capability allowing children from different cultural backgrounds to use our application to read and listen to stories. Finally, we will be deploying a recommendation system that will suggest stories to our users based on their individual preferences, interests, and reading history.

2.4 Goals and Objectives

The list of goals and objectives decided keeping in mind FictionGenie will be listed below. These goals will serve as a foundation for developing an AI-powered storytelling application for children. The goals and objectives are as follows:

- Developing a creative, user-friendly, and attractive application to captivate children's interest.
- Transforming traditional storytelling by using AI to generate personalized stories based on user preference with content filtering according to user age.
- Providing a huge range of recommended stories and the ability to listen to stories in multiple languages.
- An attractive user interface creating dynamic images based on story to reshape the art of storytelling.
- Parent will be allowed to view their children's insights like filtering content and setting usage limits to ensure full parenatl control.

2.5 Project Scope

FictionGenie aims to establish an AI-based story-generation application for children with multiple advanced features as enlisted in our goals. Our project's scope will include the following phases.

- We will be creating an interactive user interface to attract children. Parents will be given full access to monitor their children's usage, read content, set usage limits, and apply filters for content.
- An AI-powered recommendation will be used to generate recommended stories based on user history and preferences. As the app evolves, we will continue to modify the content to cope with the demands.
- Our designed architecture will have the ability to scale up with the ongoing user needs. Our application architecture will support the addition of new features, content, and languages seamlessly.

Overall, we aim to develop an amazing storytelling application for children meeting individuals' unique tastes and interests.

2.6 Sustainable Development Goal (SDG)

FictionGenie follows all the Sustainable Development Goals (SDGs) to create a good impact on society.

2.6.1 SDG-4 (Quality Education)

FictionGenie will transform the way of learning bby introducing AI-powered storytelling for kids making it simpler for them to learn and enjoy the numerous features of our application.



Figure 2.1: Quality Education

This figure represents the targeted SDG of our FYP (Quality Education)

2.6.2 SDG-9 (Industry, Innovation, and Infrastructure)

We aim to provide an innovative infrastructure for learning while minimizing the environmental impact by using proper filtering of content and limitations by parents for their children.
Our dedication to SDG-9 demonstrates our commitment to sustainable technological advancement and infrastructure development, ensuring that every child has access to and benefits from our interactive storytelling platform.



Figure 2.2: Industry Innovation and Infrastructure

This figure represents the targeted SDG of our FYP (Industry Innovation and Infrastructure)

2.7 Business Opportunity

FictionGenie offers fascinating business options including a variety of revenue sources including subscription services, in-app purchases, and collaborations with educational organizations, can be explored. Financial prospects are expected to develop significantly as the user base grows. Additionally, FictionGenie takes advantage of the growing number of parents and educators looking for exciting and engaging material for their children. The potential market is vast and varied.

2.8 Conclusion

Finally to conclude this chapter, FictionGenie is a bold initiative to reimagine storytelling for children in the digital age. Our initiative combines traditional storytelling expertise with the transformative potential of artificial intelligence to create a world where storytelling is dynamic, interactive, educative, and highly engaging.

Chapter 3 Literature Review / Related Work

This chapter looks at some of the earlier work that has been done in relation to our project and investigates their techniques. Currently, there are a few related applications that are based on AI-based Storytelling, out of which some important ones are discussed in this chapter. However, there is no application currently that provides features like Story generation, interactive user interface, multilingual support, text-to-speech functionality, and story recommendations on the basis of user preference in a single location like FictionGenie.

3.1 Definitions and Acronyms

The Definitions and Acronyms used in this whole chapter are listed below in their sub-sections.

3.1.1 Definitions

Artificial Intelligence: Refers to the development of computer systems capable of performing tasks that typically require human intelligence, such as natural language understanding, problem-solving, and learning.

Large Language Model: A type of artificial neural network designed to understand and generate human language text. Examples include GPT (Generative Pre-trained Transformer) models.

Statistical Machine Translation: A technique in machine translation where statistical models are used to translate text from one language to another.

Application Programming Interface: A set of rules and protocols that allows different software applications to communicate with each other.

Interactive Storytelling System: Software systems that enable users to interact with and influence the outcome of a narrative or story.

C Language Integrated Production System: A tool for building expert systems, which are computer programs that simulate the decision-making abilities of a human expert.

Case-Based Reasoning: A problem-solving technique that relies on past experiences or cases to solve new problems.

Recurrent Neural Network: A type of artificial neural network designed for processing sequences of data, often used in natural language processing tasks.

AIStory: An interactive storytelling prototype mentioned in one of the research papers.

TikZ: A typesetting system often used for creating high-quality graphics, including sketches.

Hierarchical Task Network: is a planning formalism that represents complex tasks as hierarchies of sub-tasks, allowing for efficient planning and execution of tasks in artificial intelligence applications.

3.1.2 Acronyms

AI: Artificial Intelligence

LLM: Large Language Model

SMT: Statistical Machine Translation

API: Application Programming Interface

ISS: Interactive Storytelling System

CLIPS: C Language Integrated Production System

CBR: Case-Based Reasoning

RNN: Recurrent Neural Network

HTN: Hierarchical Task Network

SDG: Sustainable Development Goal

3.2 Detailed Literature Review

The comprehensive literature review of prior research related to the use of AI in storytelling, story generation, and content generation is described in the sections that follow.

3.2.1 Evaluating Story Generation Systems Using Automated Linguistic Analyses

The research paper Evaluating Story Generation Systems Using Automated Linguistic Analyses is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.1.1 Summary of the research item

This research paper[1] discusses that AI-based story generation has been a rising field and a lot of research has been done and still going on but there is little research on how to evaluate these stories. The stories are usually evaluated on the basis of human feedback but the researchers identified that it's not a very useful approach and could be time-consuming in many cases as they demand repetition whenever a

new story is generated. The researchers have introduced the Automated Linguistic Analysis technique to efficiently evaluate AI-generated stories, their writing quality, and important components like sentence structure, story coherence, and grammar etc. The research work focuses on evaluating the quality of sentences generated by different models in a story continuation task. In this task, models are given an initial story and prompted to generate the subsequent sentence. The study compares sentences generated by case-based reasoning (CBR) and recurrent neural network (RNN) models with gold-standard human-authored sentences. The evaluation was performed on sentences from the Children's Book Test dataset and involved comparing the generated sentences to the gold standard. The results showed that while the gold standard sentences performed well in most metrics, differences between the CBR and RNN models were observed. The study highlights the potential of linguistic analyses in evaluating story generation and suggests their relevance in assessing and evaluating creative writing quality.

3.2.1.2 Critical analysis of the research item

The research paper has effectively explored an important research area within the domain of Natural Language Processing, mainly the evaluation of AI-generated stories using automated linguistic analysis. As the research focused on the task of story continuation, it provided valuable insights into the abilities of different models in generating feasible and contextually appropriate sentences. The study adopted an innovative approach to evaluating the content generated by AI that could be very valuable for applications like chatbots, AI-based storytelling systems, content generators, etc. The comprehensive analysis of generated sentences done in the research provides the strengths and weaknesses of different models and the comparison of AI-generated sentences with human-authored authored gold sentences offered a new benchmark to evaluate the creativity and quality of the content generated. The researchers have provided a clear methodology for their research which makes it useful for those who are targeting similar research areas. There are some notable shortcomings in the research as the research focused on using linguistic analysis to evaluate the grammar, vocabulary, and sentence structure but didn't consider the other important metrics of stories like assessing originality, novelty, and emotional impact. As the research has used only a specific data set i.e., Children's Book Test and training data from different fictional sites that could limit the findings within this domain only. A wide range of data sets could help the model more. The research lacks a deep explanation of how the comparative metrics used align with human perceptions of creativity and storytelling. While the research portrayed a solid foundation for evaluating story generation models, there is an area for expansion in terms of both evaluation metrics and applications.

3.2.1.3 Relationship to the proposed research work

The research paper discussed is related to our research of creating an AI-based Storytelling application for kids as our focus is also not on just generating stories but we aim to build stories that are effective for the readers and written in a professional way so they give a feel like the author is a human not system. The research presented in the paper offers a crucial framework for evaluating the quality of the generated text, directly aligning with FictionGenie's goal of creating AI-based stories using large language models based on user prompts. By implementing similar automated linguistic analyses, FictionGenie can ensure that the generated stories are not only creative but also linguistically and contextually robust, enhancing user engagement and satisfaction. Additionally, the research emphasizes the iterative improvement process, enabling FictionGenie to continuously enhance its storytelling capabilities based on user feedback and rigorous evaluation, ultimately delivering a high-quality and tailored user experience.

3.2.2 Interactive Storytelling: From AI Experiment To New Media

The research paper Interactive Storytelling: From AI Experiment To New Media is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.2.1 Summary of the research item

In this research paper, the problems related to the issue of lack of interest in Interactive Storytelling telling are discussed in the zone of developing Interactive Storytelling technologies. The system developed is a "Virtual Sitcom System" inspired by the characters and situation of the sitcom "Friends" implemented using the 3D game engine Unreal Tournament where the AI-based functionality of storytelling has been integrated [2]. The generation of interactive stories relies on mechanisms used for the generation of narratives. The work related to Interactive storytelling has made references to the formation of established narratives from different theories which is aligned with a theoretical framework of Computational linguistics. The approach faces challenges like the improper connection between the narrative formation and AI mechanisms that control these narrative actions, bridging the gap between diverse scriptwriting practices, such as character profiles and quasi-formal languages in the soap opera generation. To represent the roles of different characters in the Interactive storytelling system Hierarchical Task Networks (HTN) are used as they are suitable for the domain of Interactive Storytelling. These are used mainly for the purpose of structuring the behaviors of story characters and the plotlines. In the current system, the search mechanism of HTN is adopted for the real-time working of the system.

The research also discusses the challenge of staging action in a coherent manner through the introduc-

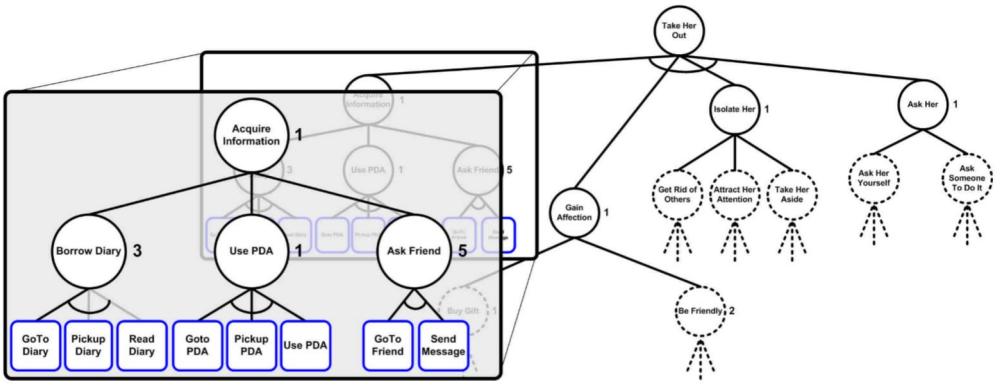


Figure 3.1: HTN Representation of lead character behaviour

This figure represents the Hierarchical Task Networks to represent the roles of different characters in the Interactive Story Telling System

tion of a Virtual Director and Cinematographer system to control the relevance of narratives and visual presentation of scenes in the established system. Research concluded that the main problem of authoring the stories needs a lot of attention and the main reason for this is that most of these working systems or prototypes of generating stories are still in research phases and have not been deeply explored or developed

3.2.2.2 Critical analysis of the research item

The research paper offers a detailed analysis of interactive storytelling and AI narration, providing detailed insights and discussing the challenges associated with it and potential solutions. It successfully delves into the complexities of narrative consistency and real-time generation using AI, showcasing the depth of understanding of the subject matter. The incorporation of planning techniques such as Hierarchical Task Networks (HTN) and the concept of a virtual director and cinematographer demonstrates innovative approaches to address the intricacies of interactive storytelling. There were some issues that the research paper left overlooked mainly the ethical considerations of integrating AI into human tasks which is a significant gap considering the ethical implications of AI technologies. The research paper lacks a discussion on the other aspects of the Ai generated narratives as it did not discuss whether the sentences that AI generated were linguistically tested or not as there could be issues related to sentence structure, coherence, and grammar. Furthermore, research could benefit from a discussion on the current limitations of AI systems in interactive storytelling, providing a more balanced perspective on the challenges faced in this emerging field. Addressing these areas would make the analysis more robust and inclusive.

3.2.2.3 Relationship to the proposed research work

The research paper discussed is highly related to our research of FictionGenie as the researchers have discussed the power of AI in generating different narratives for users while we are working on generating AI-powered stories for our readers who are basically children of different age groups. By leveraging advanced AI techniques like Hierarchical Task Network (HTN) planning that is used in the research paper, FictionGenie can enhance the depth and coherence of the stories it generates, ensuring a logical and engaging narrative flow. The fusion of established narrative formalisms, as highlighted in the research, provides a strong foundation for structuring stories within familiar literary frameworks, making the narratives not only interactive but also effective. Furthermore, the research's insights into the real-time generation of narrative actions and preserving story consistency are invaluable for FictionGenie. Implementing similar mechanisms ensures that user interactions directly influence the story's progression, creating a dynamic and immersive storytelling experience for the readers. FictionGenie can use the research's technical concepts in user-friendly interfaces, ensuring that the application remains accessible and enjoyable for its young audience while offering educational value through interactive storytelling experiences. We can also seek guidance from the research paper regarding similar issues while generating AI-based content.

3.2.3 Collaborative Storytelling with Human Actors and AI Narrators

The research paper Collaborative Storytelling with Human Actors and AI Narrators is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.3.1 Summary of the research item

This research work highlighted the use of Large Language Models like ChatGPT-3 for the purpose of AI-based storytelling to the audience [3]. The study involved using an AI system as a narrator, guiding the plot progression and character arcs while human actors performed scenes. The researchers introduced important constraints on the language model to produce longer narrative texts and tested the model in rehearsals and live performances with professional improvisers who were humans. The AI-generated narration was positively received by both audiences and performers, with the AI assisting in advancing the story and introducing unexpected plot points. The paper emphasizes the importance of human curation and timing in integrating AI-generated content seamlessly into live theater performances. While the paper does not specify the exact dataset used for training the language model, it highlights the general pre-training of GPT-3 on diverse data including books, novels, scripts of movies, newspapers, and blog posts.

3.2.3.2 Critical analysis of the research item

This paper presents a very innovative approach to using ChatGPT-3 for the purpose of storytelling by integrating it into live improvisational theater performances. The paper has highlighted many significant points related to storytelling using AI. It explores a unique area of using AI in theater-based performances for the live audience which displays the potential of AI in artistic expression and storytelling. It has shown us the collaboration between humans and AI, highlighting the creativity of AI for the sole purpose of facilitating human performers. The inclusion of feedback from both the audience and the theater performers has helped in gathering qualitative data. Moreover, the researchers have considered ethical considerations by implementing automated filters and human curation for the responsible use of AI in generating content. However, there were some notable shortcomings in the research paper. The paper lacks quantitative metrics to assess the AI-generated content. Despite the fact that qualitative feedback is valuable the quantitative data could provide a more comprehensive analysis. The paper lacked technical details like the specific prompts used, training methods, and fine-tuning techniques as these details might be useful for others who want to do similar research. The long-term sustainability of such collaborations remains unaddressed such as the questions about the novelty wearing off for both performers and audiences over time remain unaddressed. The research paper also missed the point that ChatGPT could be useful if the storytelling had to be in another language other than English. Overall, the research paper presents a unique effort in the integration of AI into theater and it it provides valuable insights into the integration of AI in artistic performances, addressing the limitations and expanding the scope of research could further enhance the depth and impact of future studies in this area.

3.2.3.3 Relationship to the proposed research work

This research paper is related to our research, as FictionGenie is also focused on using Large Language Models (LLMs). While the discussed paper used AI for storytelling purposes in the theatre, we will be using AI to generate stories and also narrate the stories for our audience which would be kids. As the paper has also considered ethical considerations, we are also focusing on that AI has to be monitored as it has to be used for kids and we have made measures so parents can also monitor what the child is doing. This paper helped our research by identifying some potential challenges and limitations of using AI to generate content that we could take care of while developing our application.

3.2.4 Story Generation from Sequence of Independent Short Descriptions

The research paper Story Generation from Sequence of Independent Short Descriptions is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.4.1 Summary of the research item

This research paper Story Generation from Sequence of Independent Short Descriptions is written by a group of members of IBM Research India [4]. It highlights how the NLG systems are weak and have limited capabilities when it comes to generating tasks requiring creativity and originality. This paper tackles the challenge of developing a coherent story from disparate descriptions. The two prominent text-generation paradigms used are SMT and DL. SMT transforms disjointed text into a cohesive narrative. Furthermore, it uses RNN which can encode variable-length descriptions into latent representations and then decode them into coherent stories. This research paper not only opens doors in generating coherent stories but also generating news articles from even related phrases.

3.2.4.2 Critical analysis of the research item

This research paper tries to produce a strong AI-based NLG system such that it will perform story generation in the same manner in terms of creativity and originality as humans. A set of approaches are used for the input of text to the AI model in this paper. PB-SMT, Syntax-SMT, and Seq2Seq are the three types of input types being discussed. PB-SMT divides the input into short phrases whereas Syntax-SMT is based on the idea of translating syntactic units rather than single words. For Seq2Seq we use multiple types varying the number of hidden layers. For the evaluation of stories generated by AI, the paper uses BLEU-4, METEOR, TER, and ROUGE-L evaluating mechanisms. The data set used is “Text Annotations” from the recently introduced Storytelling Dataset. The data is split into training, testing, and validation. After the model was trained, the results we gathered and analyzed with the evaluation mechanisms mentioned earlier. The results of the seq2seq model suffered the most because of the embedding-based approach being used. The training of the seq2seq model depicts how it saturates after every iteration and shows that the model has converged. The best-fit results we showed by PB-SMT and it outnumbered other input techniques results. These results show that Phrase-Based SMT is the best approach when it comes to generating stories using the model.

3.2.4.3 Relationship to the proposed research work

FictionGenie will be using a model for generating stories. Therefore, this research paper gives a better understanding of how the results become more perfect and coherent when using a Phrase-based approach. We will be using the genre selected by user, age, and some other inputs one by one to the model to obtain better results rather than using a whole paragraph at once as an input parameter to the model. The metrics used to evaluate the output have statistically proved the significance of the phrase-based approach. Furthermore, we can also use these metrics to check the coherence and quality of narration when testing our model.

3.2.5 Automatic Story Generation: Challenges and Attempts

The research paper Automatic Story Generation: Challenges and Attempts is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.5.1 Summary of the research item

This research paper Automatic Story Generation: Challenges and Attempts is written by a group of members of the Georgia Institute of Technology [5]. The field of automated story generation has gained a lot of attention with the ongoing technological advancements. This research paper highlights the challenges being faced during automated story generation. It also explores how previous research tackled these issues and discusses future directions to be followed now. The key challenge is generating stories that remain on topic and maintain event consistency. The two crucial points discussed in this paper are controllability and commonsense. Controllability is referred to as having human input influence on the text generation using a model whereas commonsense is a narrative system with pre-defined knowledge helping in generating coherent stories.

3.2.5.2 Critical analysis of the research item

This research paper focuses on two aspects mainly, controllability and commonsense. Controlling story generation was a significant challenge a few years back due to the limitations of neural-based story generation models. However, controllability can be achieved by a few techniques namely Reinforcement Learning, Model Fusion, Plan and Write, Generation by Interpolation, and Plot Machines. Reinforcement Learning uses a pre-trained seq2seq model with a reward shaping technique and focuses on story and goal verb frequency. Model Fusion consists of 2 levels, the first one a convolutional seq2eq model while the second one improving story quality by generating a premise. Plan and Write Framework leverages symbolic planning and explores various planning strategies to improve story coherence and quality. Generation by Interpolation uses an ending-guided approach. This approach generates stories based on the first and last sentence of the story using a coherence ranking. Plot Machines is a transformer-based system for the generation of multi-paragraph stories. It uses memory state tracking to develop coherence and control the story plot. Benchmarks and Frameworks are used for incorporating knowledge into neural language models. Benchmarks like Concept Net, Atomic, Glucose, and Social QA provide datasets for the evaluation of commonsense reasoning and understanding. Frameworks like fine-tuning pre-trained language models on datasets, analyzing story quality and incorporating commonsense knowledge graphs into model encoding. These approaches aim to improve the coherence, logicality, and grammaticality of stories generated by infusing commonsense knowledge.

3.2.5.3 Relationship to the proposed research work

This paper exhibits the proposed risks and challenges to be considered when developing our model. A story must be coherent and meaningful when it comes to generating stories using AI models. Each model and framework has its pros and cons therefore a general-purpose model can be used and fine-tuned for these tasks which will work in a much better way. For our project FictionGenie, we will be using a general-purpose model because it is a crucial aspect of storytelling that the generated content must be engaging and relatable to the audience. Furthermore, the general-purpose model has a wide range of data nodes which allow them to generate much better-quality content with a variety of variations when compared to a specific story creation model.

3.2.6 Learning to Tell Tales: A Data-Driven Approach to Story Generation

The research paper Learning to Tell Tales: A Data-Driven Approach to Story Generation is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.6.1 Summary of the research item

This research paper Learning to Tell Tales: A Data-Driven Approach to Story Generation is written by two members of the School of Informatics, University of Edinburg. Computational storytelling has sparked all over the globe because of its relevance to educational institutes [6]. This research paper proposes a data-driven approach for story generation that does not require extensive manual involvement. An end-to-end system is created which uses multiple components of the generation pipeline stochastically. The system also uses a rank-based approach where multiple stories are analyzed by evaluating whether they are coherent, fascinating, and plausible.

3.2.6.2 Critical analysis of the research item

The process of story generation operates in two phases which are creating the plot of the story and transforming it into text. This research paper proposes a data-driven approach to story generation reducing extensive manual involvement. The user will provide the topic and desired length of the story to the mode. The topic itself can be a list of nouns or a sentence. The model will then use the topic and generate several possible stories around the topic. The image attached shows how the tree is constructed for story generation. The tree encodes multiple stories, and each node corresponds to a different sentence. Stories are then generated by traversing the tree and selecting nodes with the highest score. However, there are certain limitations to this approach. The text focuses on local coherence establishment there-

fore global coherence is not guaranteed. The system relies on a knowledge base, and it has limits to its adaptability on diverse topics and domains, deteriorating the system's performance as time passes. This system primarily relies on human judges for evaluation of the story which would differ from person to person. The model also poses scalability issues in which narration is difficult due to the complex structure therefore the beam search width is set to a maximum of 500 which is insufficient when generating an extensive story.

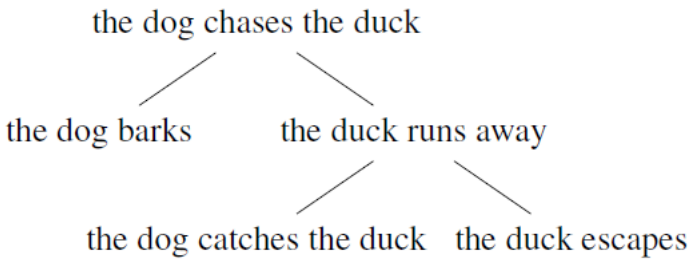


Figure 3.2: Tree Format

This figure represents how data is stored in the tree structure format.

3.2.6.3 Relationship to the proposed research work

This research paper gives an overview of the process and its limitations when generating stories. It highlights the importance of coherence by using scoring methodologies in developing a coherent story. FictionGenie can use the ranking models for assessing and selecting stories that are more likely to engage and capture children's interest. Furthermore, we can use an evaluation metric to evaluate the quality of content produced by our model. We must also keep in mind that scalability and adaptability aspects are important for an application to grow and must be kept in mind when generating a story.

3.2.7 Development of an API to Create Interactive Storytelling Systems

The research paper Development of an API to Create Interactive Storytelling Systems is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.7.1 Summary of the research item

The research paper "Development of an API to Create Interactive Storytelling Systems" was written by researchers from Ciudad University, Mexico. It covers how important an API is and its role in interactive storytelling (ISS) [7]. The research highlights the ISS topics like 3D graphics, animations, simulations, and text-to-speech which are covered by the API. The researchers have discussed multiple

storytelling models like linear branching stories, story-driven engagement, and stories generated by intelligent agents.

3.2.7.2 Critical analysis of the research item

The research study takes a thorough approach to resolving the issues associated with constructing interactive storytelling systems. Several crucial points in this research demand a close examination. The emphasis on object-oriented and modular API design throughout the article is praiseworthy. This method allows developers to concentrate on certain parts of ISS development rather than worrying about low-level implementation concerns. This modularity is consistent with best practices for modern software development. The paper analyzes various models of engaging storytelling, emphasizing the differences between user flexibility and story quality. However, it falls short of offering precise advice on which model is best suited for various cases. More in-depth study and practical suggestions in this area would have been beneficial. The incorporation of capabilities such as 3D graphics, physics simulation, and text-to-speech translation within the API is innovative. These technologies improve the user's immersion in ISS. However, the article might have gone into further detail about the obstacles and trade-offs that come with integrating these technologies.The report outlines ambitions to integrate a rule-based framework based on CLIPS for more complicated agent behavior. While this is a promising trend, further research is needed to determine how such a system can improve the standard and effectiveness of storytelling.

3.2.7.3 Relationship to the proposed research work

Several similarities exist between our proposal and the suggested research work outlined in the article. Both projects aim to improve the level of interaction and standard of story-telling experiences. Our project, like the research work, includes the creation of an API, however for a different reason. While the study focuses on ISS, our initiative intends to establish a storytelling and reading platform powered by AI. Both projects understand the value of API modularity and user-focused design. Both programs highlight the importance of effectively engaging users. Our initiative aims to empower children through fascinating storytelling, while the research study examines several creative storytelling methods and their effects on user interaction. As described in the article, the integration of technologies like 3D graphics, animations, and text-to-speech conversion corresponds with our project's goal of developing an aesthetically beautiful and engaging user experience. Future work for the study includes the construction of a rule-based system for agent behavior. Our study also investigates AI algorithms for personalized learning suggestions, which could benefit from rule-based systems that adapt to changing material.

3.2.8 Design implications of generative AI systems for visual storytelling for young learners

The research paper Design Implications of Generative AI systems for Visual Storytelling for young learners is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.8.1 Summary of the research item

Han and Cai's (2023) research item "Design Implications of Generative AI Systems for Visual Storytelling in Educational Settings" explores the design considerations and impacts of using generative AI systems in educational contexts, with a particular emphasis on storytelling through imagery [8]. This research aims to gather feedback from parents, teachers, and AI researchers on the storytelling application for children. The design, accessibility, and user interface are taken into account. The possible benefits and challenges of the storytelling application are discussed in detail.

3.2.8.2 Critical analysis of the research item

Han and Cai's research is an essential addition to this AI world as it addresses the implications of embedding generative AI into learning contexts. Several important points are taken into account in this study. To begin with, this research takes into consideration the feedback of parents, teachers, and AI professionals. It also considers that using AI tools in education involves multiple points. Secondly, the cons of AI's impact on children's decision-making capabilities are discussed. The significance of AI-powered tools outnumbers their cons also they adhere to ethical standards and guidelines when it comes to generating visuals for children. User-friendly interface and domain-driven features are based on user-centered architecture. The AI models should be fine-tuned to address safety concerns and discrimination. These key features are necessary for an AI system to get a reliable educational experience. The AI story prototype presented in the research presents the actual research findings.

3.2.8.3 Relationship to the proposed research work

We aim to develop an AI-powered tool for story generation and it is closely linked to Han and Cai's research. The work done in this research offers an insight into the whole story generation process helping us plan on developing our project. Our project aim is to prioritize a user-centric approach by focusing on the needs of children. This will ensure that the end product is attractive and will engage the audience. We will be following the safety and ethical dilemma in our system which incorporates filtering and removing biases in content. We will be taking feedback from parents, teachers and other

experts to make sure our application serves its purpose.

3.2.9 Ambient Adventures: Teaching ChatGPT on Developing Complex Stories

The research paper Ambient Adventures: Teaching ChatGPT on Developing Complex Stories is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the sub-sections below.

3.2.9.1 Summary of the research item

The research paper ”Ambient Adventures: Teaching ChatGPT on Developing Complex Stories” was written by Zexin Chen, Eric Zhou, Kenneth Eaton, Xiangyu Peng, and Mark Riedl. All of them are from the Georgia Institute of Technology. The concept of story generation and its application is being discussed in this research [9]. The project aims to generate large/complex stories using large language models (LLMs). This involves using real-world stories as a base to create fictional storylines so that children can have an interactive storytelling experience. The researchers make use of a text adventure game as a testbed to examine the agent's capacity to engage in imaginative play, replicating a house where the agent can interact depending on the generated stories.

3.2.9.2 Critical analysis of the research item

The study contributes significantly to the domains of interaction between humans and artificial intelligence and it has provided us with thorough knowledge about how we can use the knowledge of real-world stories as a foundation for FictionGenie's stories so the users will have experience of reading stories to which they can feel as being written by a human no an AI. This paper proposes the concept of imaginative play as a technique for improving human-AI interaction. The authors hope to make AI agents more appealing and approachable to users, particularly youngsters, by allowing them to engage in creative play. This innovative approach provides a fresh viewpoint on how AI might be used in everyday situations. Additionally, the study employs large language models, such as Google's Gemini, to generate stories. This method demonstrates LLMs' versatility beyond text production for storytelling purposes. The study does, however, acknowledge limitations, such as the requirement for structured cues and difficulties in understanding interactive activities in reality. Interestingly, text adventure games are used as an experiment for evaluating the agent's capacity to engage in imaginative play. It offers a controlled environment in which to evaluate the usefulness of generated stories in guiding the agent's interactions. This method bridges the gap between story creation and real-world implementation. The research report recognizes limitations, such as the requirement for explicit prompts, difficulties recognizing linkages between objects, and the model's tendency to provide recurring or lengthy descriptions.

These challenges identify areas for future development.

3.2.9.3 Relationship to the proposed research work

FictionGenie has an obvious link to the research described in this article. Both projects are focused on using AI, specifically large language models, for creativity purposes. Our project leverages large language models for story generation. While the study uses story generating to encourage imaginative play, FictionGenie employs AI to help users with creative story generation and reading activities. Additionally, the research concept of engaging users through imaginative play matches FictionGenie's goal of encouraging creativity and storytelling among users. Both programs seek to increase user engagement with AI-powered content. Model Understanding issues: The research issues connected to the model's understanding of interactive activities and the necessity for structured prompts are consistent with our project's considerations. In FictionGenie, we also need to effectively direct users to reach desired goals by using structured prompts.

3.2.10 Controllable Text to Image Generation with GPT-4

The research paper Controllable Text to Image Generation with GPT-4 is studied and analyzed. The summary, analysis, and relationship of this research article with FictionGenie are discussed in the subsections below.

3.2.10.1 Summary of the research item

This research paper Controllable Text to Image Generation with GPT 4 is written by a group of members affiliated with UC Berkeley and Microsoft Research. This research paper introduces Control GPT, an approach to improve text-to-image conversion using GPT 4 [10]. GPT 4 is an LLM and has excelled in generating programmatic sketches in TikZ. The sketches combined with diffusion-based models like ControlNet combine to produce realistic images. Overall research was detailed and has touched many key points however it has some associated challenges. The challenge faced in this research is the lack of datasets having aligned text, images, and sketches to train our model. To solve this problem, authors convert instant masks from existing datasets into polygons and eventually simulate the sketches used during testing. Control GPT provides controllability over image creation, hence providing outstanding results in spatial arrangement and object positioning. This research leverages LLMs to unleash the performance in computer vision tasks.

3.2.10.2 Critical analysis of the research item

Precisely controlling image generation from text input is a formidable challenge. Models like DALL-E2 or Stable Diffusion provide unsatisfactory results when specifying the size, position, and shape of the object in the image. To deal with this, extensive prompt engineering or sketches created manually are used. However, still, they are ineffective and difficult to generalize. On the other hand, models like GPT-4 harness LLMs to enhance the controllability of text-to-image generation models. Control GPT generates TikZ sketches and uses them as reference points for diffusion models. Control Net is a diffusion model that takes additional input conditions to transform sketches into realistic images. Control Net is fine-tuned by using a set of data with captions, images, and polygon sketches. The precision of Control-GPT relies on the precision and controllability of LLMs generating sketches. Controlled GPT-4 produces the best results on object positioning when prompted with dimensions, and position. Furthermore, Control GPT defeats DALLE-E2 and Stable Diffusion when it comes to generating multiple objects in a single image. The potential impact of AI agents can be the generation of malicious content or disinformation.

3.2.10.3 Relationship to the proposed research work

The above-discussed research is related to FictionGenie in a way that we will be producing dynamic themes and want to shape the way of storytelling by providing an interactive user environment. Studies show that children can interpret more quickly in terms of images and can memorize things easily. FictionGenie will be using a Controlled GPT4 for image creation as it has the most accuracy when compared with other existing text-to-image models. Furthermore, the TikZ sketches provide accurate positioning of objects and with the help of prompts can position them as needed. Controlled GPT-4 also performs outstandingly when generating images having multiple objects due to its high performance. By using structured prompts and sketches, will help our application develop accurate images for an endless user experience.

3.3 Literature Review Summary Table

In this section, we have summarized the research which we have done through a literature review in the form of a table. We have discussed the method, results, and limitations of each research paper briefly in this table. Following is the table providing findings of past articles related to FictionGenie.

Table 3.1: Summary of Literature Review

This table shows the summary of the review of literature reviews

No.	Author	Method	Results	Limitations
1.	Melissa Roemmele, Andrew S. Gordon, and Reid Swanson [1]	Evaluation of AI-generated stories based on quality sentences.	Provided a benchmark for evaluating the creativity and quality of AI-generated content.	A specific dataset was used which may not fully represent the diversity of storytelling genres.
2.	Marc Cavazza, Fred Charles [2]	HTN used to represent the roles of different characters in storytelling.	Enhance storytelling with HTN for structured characters and plots.	Neglects ethics in incorporating AI into human tasks.
3.	Boyd Branch, Piotr Mirowski, Kory Mathewson [3]	Integration of LLMs for the purpose of collaborative storytelling.	AI human collaboration demonstrates AI's potential in storytelling	Relies on qualitative feedback without providing quantitative metrics.
4.	Priyanka Agrawal and Parag Jain [4]	PB-SMT, Syntax-SMT, and Seq2Seq are used.	Best-fit results were shown by PB-SMT.	Metrics have flaws, leading to lower scores.
5.	Amal Alabdulkarim, Siyan Li, Xiangyu Peng [5]	Enhance story quality using common-sense and other techniques.	Combine all of these different approaches to improve stories.	No methods can ensure the generation of reasonable and coherent stories.
6.	Neil McIntyre, Mirella Lapata [6]	Generator constructs several stories by consulting a knowledge base.	Stories with maximum scoring are selected while the rest are pruned.	Detailed stories not created and relies on human judges for evaluation.
7.	Enrique Larios, Jesús Savage [7]	Creating a flexible API for building ISS.	Analysing various storytelling models, trade-offs between story quality.	Does not offer precise advice on which storytelling model is best suited.
8.	Ariel Han, Zhenyao Cai [8]	Applying AI for education, focusing on using images to tell stories.	AI tools promoting co-creation and focusing on safety and ethical use.	Analysis of the strategies and methods for mitigating bias effectively.
9.	Zexin Chen, Eric Zhou, Kenneth Eaton[9]	Large Language Models used to generate complex stories.	Stories needed revisions, but final versions were coherent and meaningful.	Needs clear prompts; tends to give repetitive or lengthy descriptions.
10.	Tianjun Zhang, Yi Zhang, Vibhav Vineet[10]	Control GPT by querying GPT-4 to write TikZ code.	GPT-4 excels in creating TikZ sketches and grounding tokens.	Potential impact of creating malicious content.

3.4 Related Work

The field of AI is progressing day by day with plenty of research being done in various domains of it. In this section, we will discuss some of the related applications that are based on AI-generated content. There are many applications that use AI for effective content generation but currently, there is no application available that provides dynamic story generation with AI along with image creation based on the story content, parental guidance, multi-lingual content, and text-to-speech functionality all under one platform like our application FictionGenie.

3.4.1 AI Story Generator and Maker

The AI Story Generator and Maker logo and details are discussed below.



Figure 3.3: AI Story Generator and Maker

This figure shows the logo of Mobile Application, AI Story Generator and Maker

3.4.1.1 Summary of the related work

AI Story Generator and Maker, a mobile application available on the Google Play store. This application is a product of the "Conquer App"[11]. It has features like the creation of AI-based stories on the basis of user prompts related to predefined genres.

It offers users the functionality they choose one of the predefined genres like thriller, mystery, horror, etc. and the user can add a few keywords on the basis of which the application generates a story for them. The application encourages users to buy its Premium versions that include additional features like Unlimited Story ideas, No Ads, Build Character, and Plot Twists at PKR 10,300 Yearly.

3.4.1.2 Critical analysis of the related work

This app focuses just on text-related content and does not focus on user experience as at every checkpoint there is an ad loading at the free version which affects user experience. Moreover, this application generates only textual content with no images of a character or scene from a story and no progress tracking. It generates content only in the English language and is not suitable for those who are not proficient in English and there is no concept of narration in a story like text-to-speech feature even in the paid versions of the application.

3.4.1.3 Relationship to the proposed related work

In using AI to create stories based on user inputs and predetermined genres, the ”AI Story Generator and Maker” by Conquer App, accessible on Google Play, is comparable to FictionGenie. Contrary to FictionGenie, it does not have graphics, progress tracking, or narrating capabilities, focusing only on text-based material. Additionally, the free version’s frequent advertisements degrade user experience, and it only supports English, which limits accessibility for non-English users. As a more flexible and user-friendly alternative for young readers, FictionGenie provides a more full and interactive narrative experience, featuring graphics, linguistic support, and parental controls.

3.4.2 Story Spark

The Story Spark logo and details are discussed below.



Figure 3.4: Story Spark

This figure shows the logo of Web Application, Story Spark

3.4.2.1 Summary of the related work

Story Spark, is a web application that uses AI to create personalized stories for children of multiple age groups [12]. The user inputs some story idea and the application will generate a story based on it “Story Spark”, is a very high-end application for effective story creation with a very interactive user interface. It works on the theme of generating multi-lingual stories for kids mainly. It has some pre-built stories in it and also It enables the user to first input some story ideas, the user will further have the privilege to give advance inputs like choosing the age group, pages of story, and message to convey through the story. These key functionalities are available with the payment plans i.e., 15.83 Euros or PKR 4660, per month.

3.4.2.2 Critical analysis of the related work

This application is a very powerful advancement in the field of AI-based storytelling but it provides stories only in 9 different languages, and the most of key features are paid so a user has to spend money to gain access to the majority of these features. Despite being focused on kids’ storytelling and being the most developed application in this domain, there is no focus on parental monitoring or progress tracking

in this application.

3.4.2.3 Relationship to the proposed related work

Both ”Story Spark” and FictionGenie use AI to customize children’s stories, although they have very different approaches. Although ”Story Spark” has an interactive interface and charges a price for users to input tale ideas and preferences, this limits the number of people who can use it. However, it lacks robust parental monitoring and progress tracking capabilities despite offering stories in nine different languages.

In comparison, FictionGenie stands out by providing a user-centric approach, giving parental controls, language assistance, and a wide variety of genres without charging extra for essential services.

3.4.3 Kidstory: AI bedtime stories

The Kidstory: AI bedtime stories logo and details are discussed below.



Figure 3.5: Kidstory: AI bedtime stories

This figure shows the logo of Mobile Application, Kidstory: AI bedtime stories

3.4.3.1 Summary of the related work

“Kidstory: AI bedtime stories”, is a mobile application developed by SRC Software and is available on Google PlayStore with more than a thousand downloads. This application revolves around the idea of creating stories essentially for kids [13]. This mobile application generates a story for users on the basis of three questions How many protagonists are in the story, what are its protagonists and the main input is actually the story title which is the main idea around which the whole story revolves. It generates unique stories on the basis of these inputs and the user can generate only three free stories for additional stories the user is asked to pay for credits to generate stories e.g., 10 credits in PKR 44.5. The application has some pre-built stories for users sorted according to different genres.

3.4.3.2 Critical analysis of the related work

This application generates stories for only kids with no focus on the age groups of kids. It generates text-only stories with no visuals, and all the stories are in English. Moreover, the application is flooded

with ads that affect users in the free mode. It's a very generalized application that generates a story on the basis of a single title instead of keywords. It lacks features like text-to-speech functionality, effective story creation, interactive UI, no parental support or progress tracking.

3.4.3.3 Relationship to the proposed related work

Both "Kidstory: AI bedtime stories" and FictionGenie seek to deliver AI-generated stories for children, although their methods and selections are very different. Both programs employ user input to generate stories, however "Kidstory" limits users to three free stories and charges for additional ones, whereas FictionGenie offers a more encompassing experience by include necessary features without additional fees. While FictionGenie provides language support and an aesthetically pleasing interface, "Kidstory" generates text-only stories in English. This improves accessibility and user experience. Another unique feature of FictionGenie is its text-to-speech capability, which enables interactive and interesting story-telling experiences

3.4.4 Story Seed

The Story Seed logo and details are discussed below.



Figure 3.6: Story Seed

This figure shows the logo of Mobile Application, Story Seed

3.4.4.1 Summary of the related work

"Story Seed", is a mobile application developed by "StorySeed.ai" available on App Store and Play Store. This application is designed to provide kids with stories according to different themes, and tones and on the basis of ideas given to it by the users[14]. It has more than five thousand downloads on the Google Play Store. This application allows users to create stories with the power of AI by going through three phases. Firstly, users will select the genre of the story, then they select the tone and lastly, they are enabled to give the idea of the story. It has some pre-built stories and some pre-built characters that a user can choose to be a part of their story. It has parental features and is available in 8 different languages. The stories generated by it are also comprised of images that give users a good experience.

3.4.4.2 Critical analysis of the related work

This application is a good advancement in the field of AI-driven content creation. In this application, there are different features available for users like dynamic story generation, multi-lingual support, etc. However, it has a mechanism of only story generation, but not storytelling as it has no audio concept for narrating the story to kids. The stories generated in this application are not according to the age group of kids but are very generalized i.e. same sort of story for a 4-year-old and a 13-year-old.

3.4.4.3 Relationship to the proposed related work

Both "Story Seed" and FictionGenie use AI to generate children's stories, although they have significant differences. Although "Story Seed" supports multiple languages and includes pre-made characters and visuals, it is audio-only and does not differentiate between stories for different age groups. Contrarily, FictionGenie not only creates stories but also offers age-appropriate content and text-to-speech technology for audio narration. In comparison to "Story Seed," FictionGenie gives young readers a more customized, secure, and user-friendly experience thanks to parental settings and a variety of genres.

3.5 Related Work Summary Table

Following is the table providing findings of past related work related to FictionGenie.

Table 3.2: Summary of Related Work

This table shows the summary of the review of related works

Application	**Features**	**Relevance to Application**	**Limitations**
AI Story Generator and Maker [11]	Redefined genres and provide keywords to generate AI-based stories	Based on user prompts and predefined genres	Focuses only on text-based content, lacks images or progress tracking
Story Spark [12]	Personalized stories for children based age group, pages, and story ideas	Personalized children's stories	Limited number of languages, and many key features require payment
Kidstory: AI bedtime stories [13]	User-provided information, such as the number of protagonists and story titles	Approach of generating stories for children based on user inputs	Generates text-only stories in English,
Story Seed [14]	Allows users to select story genres, tones, and provide story ideas	Feature of generating stories based on user preferences and offering visual content	Story Seed lacks an audio narration features

3.6 Conclusion

The research papers under examination offer light on the developing field of automated story development, highlighting the difficulties and creative solutions used to overcome them. Controllability, coherence, and the incorporation of common sense knowledge into story production are among these difficulties. While several methods, including model fusion, reinforcement learning, and data-driven approaches, have been investigated to improve story development, each method has advantages and disadvantages. The variety and potential of AI-driven storytelling have also been illustrated by the incorporation of AI into artistic performances and the examination of ethical issues. These realizations are extremely important to our proposed study with FictionGenie, where we want to use AI to produce interesting, cogent, and context-aware children's stories while taking ethical considerations and quality assessment into account.

3.7 Selected Model

We have decided to use a Large Language Model (LLM) in our project like Gemini and ChatGpt which can now not only create stories but uses the DALLE-3 model to generate dynamic images based on the prompts provided. Models like Gemini can be customized for certain tasks making them adjust their performance for our project needs. Gemini and ChatGPT are widely used and as we know they learn more when being used therefore they are the perfect models for generating stories and images.

Chapter 4 Software Requirement Specifications

This chapter describes the important features of the project. It will include functional, non-functional requirements, assumptions, use case diagrams and multiple other subtopics involved in our project.

4.1 List of Features

FictionGenie will provide the following features to its users:

- Children will be able to read stories.
- Dynamic generation of stories based on penalization.
- The app will include a dictionary to help kids learn the meanings of new words.
- Text to Speech will allow children to listen to the stories.
- Parents will be able to view their child's usage of the application.
- System will create dynamic images based on the story context with an interactive UI.

4.2 Functional Requirements

This subtopic will discuss the functional requirements of children, parents and the system individually.

4.2.1 Functional Requirements of Children

System will allow children to:

- Generate personalized stories based on the preferences.
- Choose from various languages for story narration.
- Login to the account.
- Easy access to the dictionary to look up words.
- Listen to generated stories.

4.2.2 Functional Requirements of Parents

System will allow parents to:

- Create a new Account.
- Login to the account.

- Creating a secret pin to get child analytics.

4.2.3 Functional Requirements of System

- Generate dynamic images based on the story.
- Creating personalized stories based on user preferences.
- Filter the input parameters of the user to generate stories.

4.3 Quality Attributes

The quality attributes that FictionGenie aims to follow are enlisted below.

4.3.1 Reliability

The application 'FictionGenie' will prioritize dependability to offer a continuous and steady user experience. To achieve a reliable application, all the algorithms and recommendation engines will be validated so that reliable story suggestions are provided to the user. We will be using a strong infrastructure so that our application has zero downtime and users can have a flawless experience.

4.3.2 Maintainability

The project will ensure easy maintainability by making the codebase well-documented along with using a modular approach so that it will adhere to industry standards. Our application will be continuously updated to address any known faults incorporated. The addition of new features and upgrades will be done timely by the changing technology ensuring that our application will run for a long time.

4.3.3 Usability

Our application will be designed keeping in mind a user-friendly experience. The application will have appealing visuals, intuitive navigation, and parental controls. Users will be allowed to provide feedback and it will be used to improve the usability of the application.

4.3.4 Correctness

It is crucial to ensure that the recommendation engine and age-appropriate content are working properly. The AI algorithms used in our application will be tested completely to ensure all the faults and flaws are eliminated before launching the application for our users. Parental control will allow parents to ensure that their child is learning in a secure and safe digital environment.

4.3.5 Efficiency

In terms of story generation, content delivery, and application response, efficiency will be a major quality attribute. AI algorithms will be optimized for speed and resource utilization, ensuring that both children and parents have a smooth and quick experience. The program will be built to efficiently manage a large number of stories.

4.3.6 Flexibility

FictionGenie recognizes the ever-changing nature of human preferences and technological improvements. Flexibility will be built into the application's design to allow for future updates, feature expansions, and the incorporation of new languages. This adaptability keeps 'FictionGenie' at the forefront of AI-powered storytelling, matching changing user expectations.

4.4 Non-Functional Requirements

Non-functional requirements are characteristics of a system that indicate how it should function; they do not, however, outline the features and functions of the system.

4.4.1 Performance

The FictionGenie's performance will be a significant non-functional requirement. With the help of cloud computing, it should provide quick content recommendations, short load times, and smooth transitions between stories. The program should be optimized to efficiently manage multiple user interactions, ensuring a responsive experience even during peak usage periods.

4.4.2 Reliability

The application's reliability is a non-negotiable requirement. The system will ensure minimal downtime, zero crashes, and no disturbance in viewing content to ensure a reliable application. Regular security updates will be provided to ensure user data is kept securely. A team will be monitoring the application's performance at the back end and regular maintenance updates will be provided to ensure a seamless user experience.

4.4.3 Security

Our application will ensure a secure user environment. All of the user's details will be stored in the database by using encryption routines. Data confidentiality will be our top priority. In case of a security breach, we will adhere to the international standards of informing our users regarding it.

4.4.4 Compatibility

Compatibility is an important aspect ensuring that every single individual must have a flawless experience when using our application. We will be ensuring that our application will be supported and compatible with all devices.

4.5 Assumptions

The following assumptions have been made for the application:

- Users will have uninterrupted access to the internet.
- Users will have adequate storage capacity for applications to be installed.
- We are assuming that all users of the application have basic knowledge of how to use a phone.

4.6 Use Cases

This section lists the use cases used in our application

4.6.1 User Registration

Name	User Registration		
Actors	Children and Parents		
Summary	The user shall provide their necessary information and after that user will be registered.		
Pre-Conditions	User has a valid email address and the user is not already registered.		
Post-Conditions	The user's registration details are stored in the database.		
Special Requirements	Captcha verification must be used to prevent auto registration		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the registration page.	2	Registration page asks for email, password, dob, and secret pin.
2	The user enters all valid details.	4	The user gets logged in and can now generate stories.
Alternative Flow			
3	The user enters invalid registration details.	4-A	The system responds with an error message: Incorrect details.

4.6.2 User Login

Name	User Login		
Actors	Children and Parents		
Summary	The user shall provide their necessary information and after that user will be logged in.		
Pre-Conditions	User has a valid email address and already registered		
Post-Conditions	The user is granted access to the application.		
Special Requirements	Captcha verification must be used to prevent bot attack		
Basic Flow			
Actor Action		**System Response**	
1	The user navigates to the login page.	2	System will ask for user email and password.
2	The user enters email and password.	4	After the email and password are verified from the database user is allowed to read stories.
Alternative Flow			
3	The user enters invalid login details.	4-A	The system responds with an error message: Incorrect details.

4.6.3 Story Generation

Name	Story Generation		
Actors	Children		
Summary	The children will provide a set of details to generate a personalized story.		
Pre-Conditions	The user is logged into FictionGenie. The user is on ”Story Generation” page.		
Post-Conditions	The system generates a unique story based on the user’s input parameters.		
Special Requirements	Captcha verification must be used to prevent bot attack		
Basic Flow			
Actor Action		**System Response**	
1	User selects ”Story Generation” feature.	2	The system displays a form prompting the user to input genre, age, and the desired difficulty level for the story.
2	User inputs(genre, age, difficulty level).	4	The system validates the input parameters.
3	Input parameters used to generate Story.	6	The generated story is composed real-time, incorporating input and displayed to user.
Alternative Flow			
3	User provides incomplete or invalid input.	4-A	The system prompts the user to correct or complete the required fields before proceeding with story generation.

4.6.4 Parental Control

Name	Parental Control		
Actors	Parent/Guardian		
Summary	Parents access the parental control interface to regulate and monitor their child's interaction with the application.		
Pre-Conditions	The Parent is logged into FictionGenie. Parental Control is enabled using secret password		
Post-Conditions	Parent gains access to children's activities. Content restrictions applied by parent's.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Parent goes to "Parental Control" and enters pin	2	System verifies pin and grants access Parental Control settings.
2	Parent views insights of their children's.	4	System displays analytics of children's reading and learning progress.
3	Parent sets restrictions for children	6	The system saves the content restriction settings for the respective child profiles.
Alternative Flow			
3	Parent enters an incorrect pin.	4-A	System denies access and prompts to re-enter the secret pin.

4.6.5 Multilingual Story Selection

Name	Multilingual Story Selection		
Actors	Children		
Summary	Children can read stories in different languages as selected by them.		
Pre-Conditions	The User is logged into FictionGenie.. User is on Story Generation page.		
Post-Conditions	Story is presented to the User in the chosen language.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The User selects "Language Type" option.	2	System provides a dropdown or with various language options.
2	User selects Language Type and provide input parameters of story	4	System loads the chosen Story in the selected language.
Alternative Flow			
3	User cannot find desired language option.	4-A	System provides an option to request additional languages

4.6.6 Text-to-Speech

Name	Text-to-Speech		
Actors	Children		
Summary	After the story is generated, the Children can listen to the story by pressing the text to speech button.		
Pre-Conditions	The User has generated a story within FictionGenie. Story content is displayed on the page.		
Post-Conditions	System converts the text into speech. User can listen to the generated stories.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	After Story generation, user gets Text-to-Speech option.	2	A button is displayed, indicating the availability of Text-to-Speech.
2	User clicks ”Text-to-Speech” button.	4	System initiates the conversion of written story into speech.
3	System generates speech corresponding to the story.	6	The Story audio is played automatically.
Alternative Flow			
3	Text-to-Speech conversion encounters an error.	4-A	System provides an error message.

4.6.7 Dynamic Image Generation

Name	Dynamic Image Generation		
System	Children		
Summary	After a story is generated, FictionGenie dynamically generates images based on the story content to enhance the user interface.		
Pre-Conditions	User has generated a story in FictionGenie. The story content is available.		
Post-Conditions	System dynamically generates and integrates images into the user interface.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	After Story generation, system analyze generated story content.	2	System identifies key elements, themes, or scenes within the story.
2	Generated Images integrated into the UI.	4	System incorporates dynamically generated images into UI, ensuring visually appealing design.
Alternative Flow			
3	System is unable to generate images.	4-A	A Static general image is displayed.

4.6.8 App Interface Dark/Light Mode

Name	App Interface Dark/Light Mode		
System	Children		
Summary	Users can switch between Dark and Light modes based on their preferences for readability and visual comfort.		
Pre-Conditions	User has FictionGenie app and is logged in.		
Post-Conditions	App interface reflects the selected Dark or Light mode.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User selects the ”Dark Mode” or ”Light Mode” option.	2	System updates app interface to reflect the chosen mode.
Alternative Flow			
3	User has set app to follow device’s appearance settings.	4-A	App interface automatically adjusted to Dark or Light mode based on the device settings.

4.6.9 Story Recommendation

Name	Story Recommendation		
Actors	Children		
Summary	FictionGenie saves previous user preferences, providing personalized story recommendations.		
Pre-Conditions	The User is logged into FictionGenie. User is on Story Generation page.		
Post-Conditions	User preferences and reading history are updated for future recommendations.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	System presents a list of recommended stories.	2	App displays story recommendations based on the user’s history.
2	User selects a recommended story to read.	4	System loads the chosen story, ensuring a personalized reading experience.
Alternative Flow			
3	System has Insufficient data to generate recommendations.	4-A	App does not provide any suggestion, user has to input values.

4.6.10 Instant Word Definition

Name	Instant Word Definition		
Actors	Children		
Summary	Users can access word meanings within the app by pressing and holding on a word, to get definitions.		
Pre-Conditions	The User is logged into FictionGenie. User is reading a Story.		
Post-Conditions	System displays the meaning of the selected word.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User is reading a story within app.	2	App displays the text with words that can be selected for definition.
2	User presses and holds on a specific word.	4	Integrated Dictionary triggered, providing pop-up with meaning word
Alternative Flow			
3	User accidentally triggers dictionary while scrolling	4-A	App dismiss pop-up for inadvertent activations.

4.6.11 Add New Story in the Application

Name	Add new stories into the application for the user		
Actors	Admin		
Summary	Admin can add stories with images into the application that can be read by the children		
Pre-Conditions	The user is logged in as an admin and is on add story panel.		
Post-Conditions	After admin has added text of the story with images and other parameters, this story will be visible to all the children		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Admin clicks on the add new story button.	2	App displays a form with multiple input fields.
2	Admin enter text, genre, images and other parameters and clicks on ADD.	4	The story will be added to the system and children can now read it.
Alternative Flow			
3	Admin entered wrong inputs for the story	4-A	App also shows the story in preview mode to the admin before going live.

4.6.12 Read Pre-built Stories

Name	Read pre-bult stories		
Actors	Children		
Summary	Children can read pre-built stories which are already available in the application.		
Pre-Conditions	The user is logged in as a child and can see the thumbnails of the pre-built stories.		
Post-Conditions	Children can now see the text with images of the already generated stories.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Child select specific genre to read stories.	2	App displays all the pre-built built stories releted to that genre.
2	Child is logged in and select on of the pre-built stories to read.	4	App displays the the story with text and images.
Alternative Flow			
3	User is not logged in.	4-A	App will redirect the user to login screen.

4.6.13 Update Profile

Name	Update user profile		
Actors	Children and Parents		
Summary	Users can update their profile like profile picture, profile name, reset passwords, etc.		
Pre-Conditions	The User is logged into FictionGenie. User is on My profile screen.		
Post-Conditions	Update user profile will be displayed to the user after changes.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User clicks on My Profile	2	App displays the user profile with its details in the input fields.
2	User updates the information or reset password.	4	System will update the user profile and new profile will be displayed to the user.
Alternative Flow			
3	User forgot his/her old password.	4-A	System will redirect to user to the forgot password procedure.

4.7 Hardware and Software Requirements

The following subsections list the hardware and software requirements that are essential for the application to be developed and deployed.

4.7.1 Hardware Requirements

The hardware requirements for a phone application like FictionGenie will require the following specifications:

- Operating System: Android or IOS
- Processor: Mid Range to High end for smooth performance
- RAM: At Least 2GB
- Storage: At Least 30MB
- Display: A High-Resolution Display for a better experience if possible
- Network Connectivity: Reliable Internet Connection

4.7.2 Software Requirements

Following are the software requirements for our project:

- Flutter SDK and Dart for app development
- Integrated Development Environment Visual Studio for coding
- Python with Flask for backend development
- Google Firebase for storing data
- AWS Lambda for serverless functions
- Integrating Google Gemini, Chat GPT , Voice to Speech and Dictionary API
- GitHub will be used as the version control system

4.8 Graphical User Interface

This section should give the GUI dumps of each screen, concerning the users. The navigation flow of each user is also required, and each GUI should mark the functionality/use case that it covers. We have designed these screens using Figma a famous tool of UI/UX design and the device on which the GUI is being shown is iPhone 14 pro max and Google Pixel.

4.8.1 Welcome Page

This page opens up as the welcome page of our application when the user opens the application.

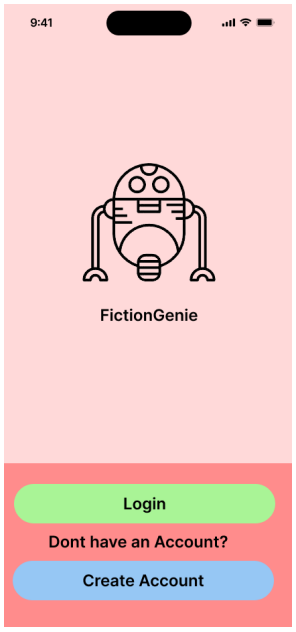


Figure 4.1: Welcome Page

This figure shows the Welcome Page of our Application

4.8.2 Login Page

This page opens once the user has decided to log into the application

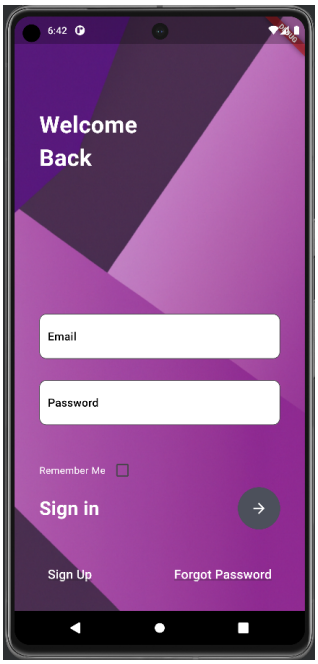


Figure 4.2: Login Page

This figure shows the Login Page of our Application

4.8.3 Signup Page

This page opens once the user has opted to create account on FictionGenie



Figure 4.3: Signup Page

This figure shows the Signup Page of our Application

4.8.4 Forgot Password Page

This page opens if a user has forgotten his/her password and wishes to change it.

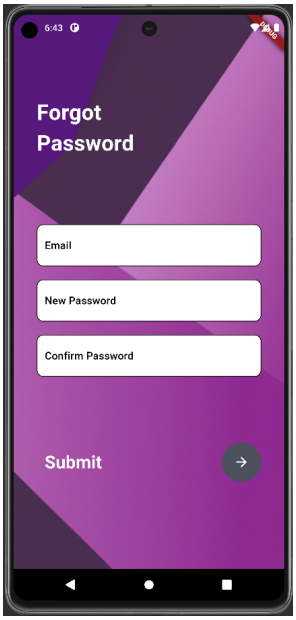


Figure 4.4: Forgot Password Page

This figure shows the Forgot Password Page of our Application

4.8.5 Profile Avatar Selection Page

This page allow the users to select Avatar as per their choice.

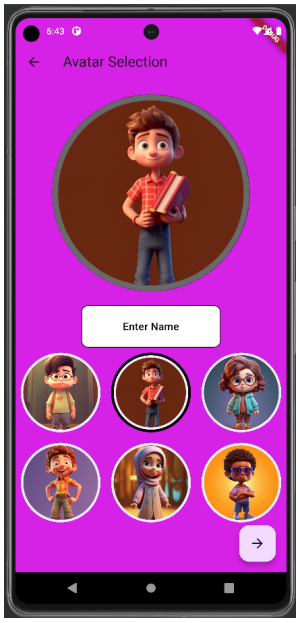


Figure 4.5: Profile Avatar Selection Page

This figure shows the Profile Avatar Selection Page of our Application

4.8.6 Home Page

This page opens once the user after the user has logged into the application and is now being presented with the menu of the application.

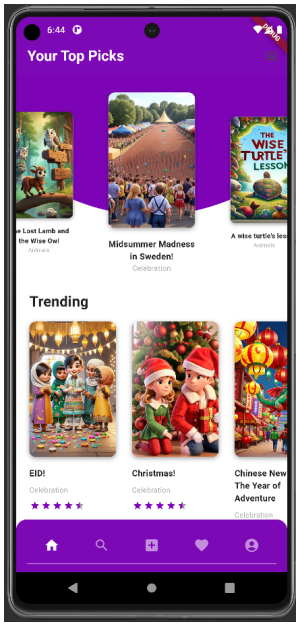


Figure 4.6: Home Page

This figure shows the Home Page of our Application

4.8.7 Story Generation Page

This page opens after the user has chosen to generate an AI-powered Story. This page prompts users to enter parameters related to the story being generated.

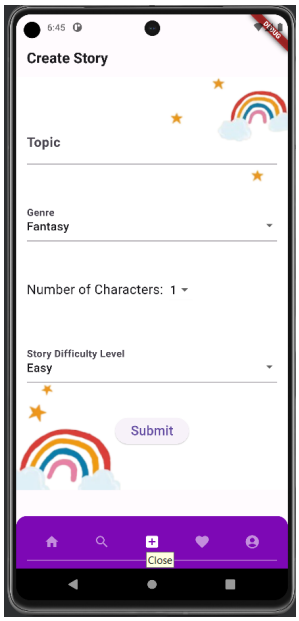


Figure 4.7: Story Generation Page

This figure shows the Story Generation Page of our Application

4.8.8 Story Page

This page opens when the user is reading a story whether it is a prebuilt story or an AI-generated story.

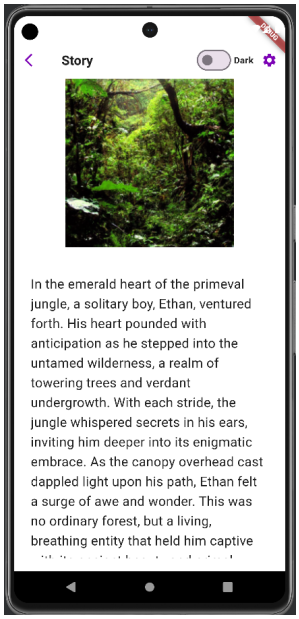


Figure 4.8: Story Page

This figure shows an example story of our Application

4.8.9 Story Page in Dark Mode

This page appears when the user is reading a story in dark mode.

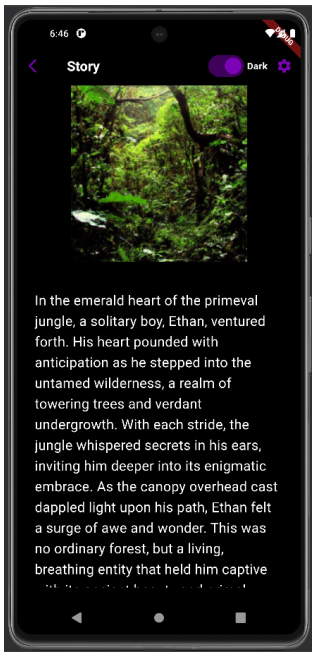


Figure 4.9: Story Page in Dark Mode

This figure shows an example story of our Application in Dark Mode.

4.8.10 Explore Story Page

This page allows users to search for stories based on different parameters like Genre, New Release etc.

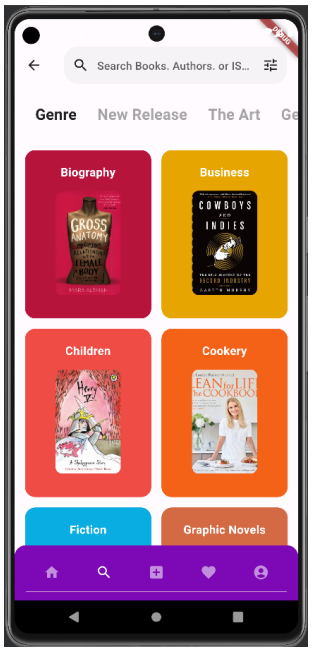


Figure 4.10: Explore Story Page

This figure shows the explore story Page of our Application.

4.8.11 Recently Viewed Stories Page

This page allows users to view the stories that they have read in the past.

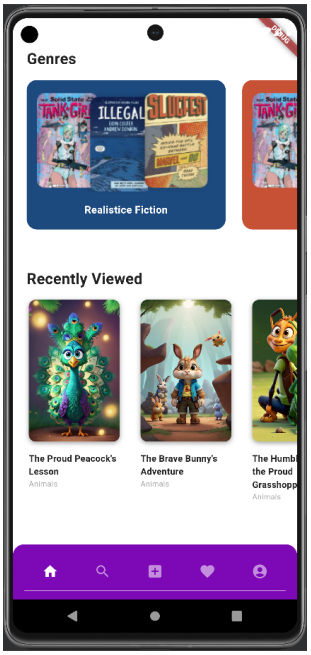


Figure 4.11: Recently Viewed Stories Page

This figure shows the Recently Viewed Story Page of our Application.

4.8.12 Favourite Stories Page

This page allows users to view the stories that they have marked as favourites.

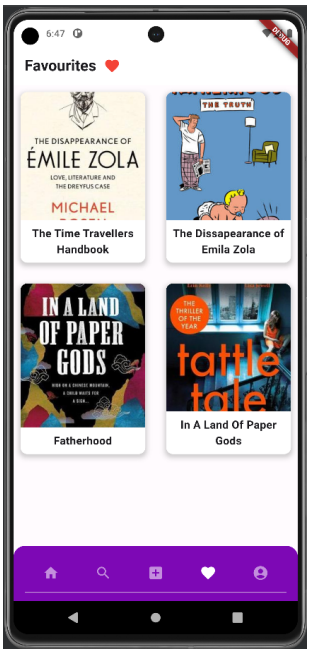


Figure 4.12: Favourite Stories Page

This figure shows the Favourite Stories page of our Application.

4.9 Database Design

A database is necessary for this project to store user details, story details, and other relevant details for the proper and synchronized operation of the application.

4.9.1 ER Diagram

The Entity-Relationship (ER) diagram for the Fiction Genie app summarizes the main elements and how they relate to one another. Users can possess several devices and interact with a variety of stories thanks to their distinct identities and related profiles. The content, genre, and age range of stories are closely related to users, interactions, and imagery. The system's architecture is comprehensively visualized in the diagram, with a focus on AI-driven, interactive, and user-centric story generation features. The ER diagram is shown below:

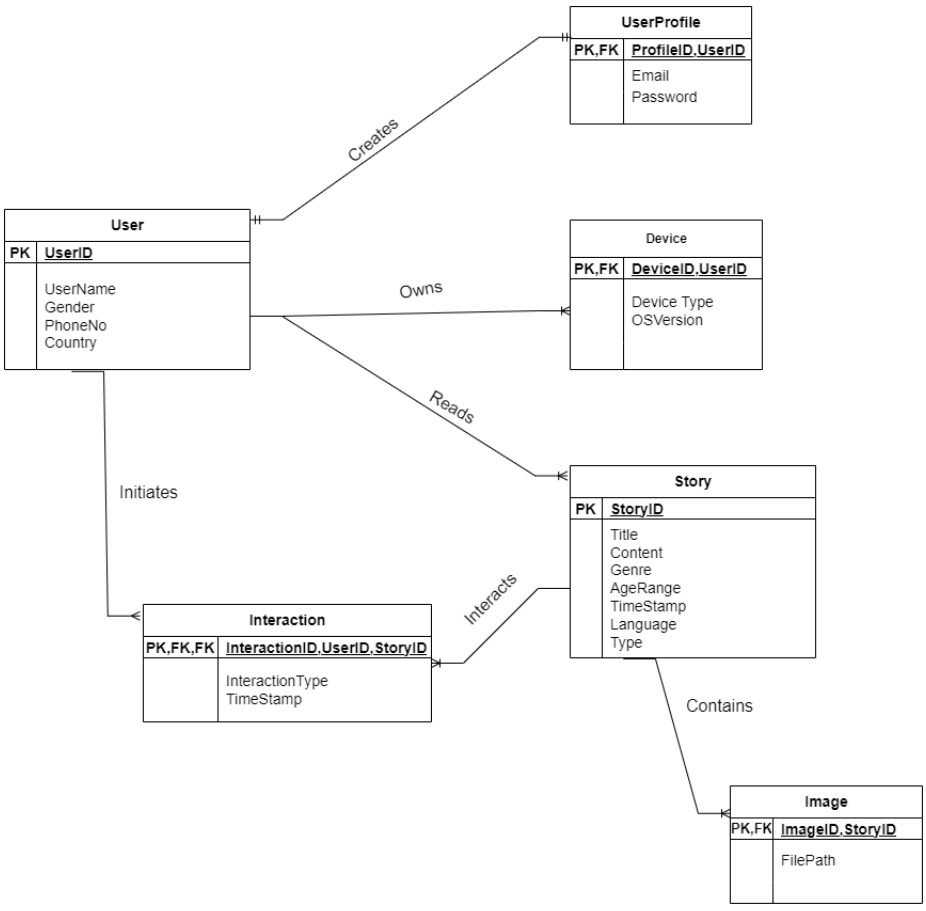


Figure 4.13: ER Diagram

This figure shows the ER Diagram related to the Database Design of our Application

4.9.2 Data Dictionary

This table contains all the entities along with their attributes in the below table:

Table 4.1: Data Dictionary Table

The given table gives the details of the tables of the database

Entity	Attributes	Data Type	Relation To	Nullable	Description
User	UserID	Integer		No	Id of the user (PK)
	Gender	String		No	Gender of the user
	PhoneNo	String		No	Phone no of the user
	Country	String		No	Country where the user lives
UserProfile	ProfileID	Integer		No	Id of the profile created by the user (PK)
	UserID	Integer	User	No	Id of the user (FK)
	Email	String		No	Email of the user
	Password	String		No	Password of the user
Device	DeviceID	Integer		No	Id of the user's device (PK)
	UserID	Integer	User	No	Id of the user (FK)
	DeviceType	String		No	Device type i.e Android or iOS
	OSversion	String		No	OS version of the device
Story	StoryID	Integer		No	Id of the Story (PK)
	Title	String		No	Title of the story
	Content	String		No	The content of the story which user will read
	Genre	String		No	Specific Genre of the Story
	AgeRange	String		No	Age Range of the Story eg (10 to 15)
	TimeStamp	Date		No	Date at which the story was created
	Language	String		No	Primary Language of the story
	Type	String		No	The type of story i.e AI-generated or Predefined

Image	ImageID	Integer		No	Id of the Image (PK)
	StoryID	Int	Story	No	Id of the story to which image belongs (FK)
	FilePath	String		No	Path where the image is stored
Interaction	InteractionID	Integer		No	Interaction between the user and the story
	UserID	Integer	User	No	Id of the user (FK)
	StoryID	Int	Story	No	Id of the story (FK)
	InteractionType	String		No	Type of interaction i.e Reading, or Generating the story
	TimeStamp	Date		No	Time of the interaction

4.10 Risk Analysis

Developing a project like FictionGenie involves multiple risks. The integration of APIs, protecting user's personal information, maintaining app performance during peak usage, and adhering to regulations is challenging. Furthermore, there is a strong need to maintain user engagement so that the application will not get the user bored. There can be service outages due to nonavailability of the internet. User details like email, password, and personal information can be stolen if not securely saved. The application may experience performance issues and could potentially encounter errors in the event of API downtime or disruptions.

4.11 Conclusion

To conclude this chapter provides an in-depth software requirement specifications linked with FictionGenie. The functional and non-functional requirements are discussed in this chapter along with software and hardware requirements. A comprehensive list of features is provided which will be developed. The Database and Use Case design provides an insight into the whole system, specifying how data will be organized and accessed, and outlining the various use cases that depict the interactions between the users and system. This comprehensive chapter provides a crucial foundation for the development and implementation phases of the application.

Chapter 5 Proposed Approach and Methodology

The proposed approach and methodology will discuss the suggested plan and systematic process involved in building the application. The integration of API along with the approach to be followed during the project life-cycle will be discussed.

5.1 Story Generation with Google Gemini API

The user story will be generated using Gemini. The user will input some parameters like age, genre, and difficulty level which will be sent to the Gemini API after filtering the inputs. Filtering is important to prevent young children from reading age-restricted content. For example, a 5-year-old child will not be able to read horror or action stories as it might have a wrong impact on his/her nature.

5.2 Agile Development

We will be deploying this application on the Google Play Store as well therefore, an agile development method will be used. The development will be done iteratively so that continuous improvement can be done to the application. The whole project will be divided into a series of sprints and regular feedback will be taken into account.

5.3 Text to Speech API

The option to listen to stories is also integrated into the application. This will allow children having difficulty reading or have partially impaired vision, to listen to the stories. Google text-to-speech API will be used to convert stories into audio files so that children can enjoy a seamless experience.

5.4 Dynamic Image Generation

Generating images dynamically will be critical in improving the application's visual attractiveness. A specialized module is created to analyze significant aspects, themes, or scenes in the generated stories. Dalle 3 is used to produce pictures that enhance the story's content on the fly. If the system has trouble generating images, a fallback mechanism will display a static general image to provide a smooth user experience. The images generated are related to scenarios of the story.

5.5 Multilingual Support

The application will have multilingual support to enhance cultural diversity and global accessibility. Children will be able to use their preferred language to read stories. A language selection module will be included, giving users the option of selecting from a number of languages. Data will be sent from this module to the best translation API to convert the story into the child's preferred language. This translated story will be displayed to the child through our best user interface.

5.6 User Centric Design

The application's user interface will prioritize user experience, with a focus on children's particular needs. An intuitive and visually appealing interface will be created as part of a user-centric design strategy. To engage and captivate the interest of young users, the application's interface will be created within the Flutter framework with vibrant colors and appealing images. The design will be continually refined depending on user preferences and usability through regular user testing and feedback loops.

5.7 Conclusion

To summarize, FictionGenie's methodology combines cutting-edge technology and user-centered design to provide a compelling and beneficial story-reading experience. The app offers a secure and enriching platform for children globally through features such as Gemini API, agile development, and ethical content filtering, and it is at the forefront of new AI-powered storytelling.

Chapter 6 High-Level and Low-Level Design

This chapter will discuss the high-level and low-level design for our application. In the high-level design, we will focus on the overall architecture, key modules, and their relationship with other components. The low-level design elaborates on the individual modules, classes, and methods. It will incorporate how data structures and algorithms will be implemented in these modules. Both the high-level and low-level design play a critical role in developing our application FictionGenie.

6.1 System Overview

FictionGenie is aimed to provide an endless user experience for children. It will allow children to engage with the application and improve their skills by reading and listening to stories. The dynamic images generated in stories will attract children to imagine the story much better. We will be applying appropriate filters to ensure children are not exposed to vulnerable content. The following modules will be included in our system:

6.1.1 AI-Based Story Generation

FictionGenie will be using a Large Language Model like Gemini to generate unique stories for children. The prompt will be given from the application sending genre, age, language, and difficulty level to the model to generate a story. The story content will also be filtered so that it is age appropriate also if any filter of content is applied by parents, the application will consider it.

6.1.2 AI-Generated Images

This module will deal with how images will be generated by the stories. FictionGenie will generate dynamic images for an interactive user interface for children. The application will use ChatGpt to create dynamic images at run time. The images generated will captivate young minds and will develop their interest in storytelling. The images will help children understand the stories much better and imagine a visual experience.

6.1.3 Pre-built Stories Library

The application will provide a huge library of pre-built stories of multiple genres. This will enable children to select stories from the available ones and read them instantly. We will be adding multiple genre stories like mystery, action, fantasy, and adventure to cater to the taste of all children.

6.1.4 Story Narration with Text-to-Speech

In this phase, with cutting-edge text-to-speech technology, the app provides smooth narration to improve the user experience making the app useful for those who are unable to read properly. Moreover, youngsters can engage in interactive storytelling and support reading skills by listening to stories aloud.

6.1.5 Multilingual Support

This phase deals with the integration of multilingual features like a translator in the application. With the integration of this module, a wide audience can utilize FictionGenie according to their language, regardless of their proficiency in English.

6.1.6 Age-Specific Content

This phase will make the app able to accommodate a range of age groups, the content that is produced is both developmentally appropriate and appropriate for the target audience. Based on the user-specified age range, the application modifies the storytelling style and level of complexity to deliver content appropriate for early childhood, pre-adolescence, and beyond.

6.1.7 Parental Monitoring

This phase will cater to the integration of parental monitoring systems in the application that is being developed mainly for young kids. This module will take care of the well-being of the children. Parents can set content filters, monitor usage statistics, and receive insights into their child's interaction with the app.

6.2 Design Considerations

In order for our application to be successful, we must address every issue and dependence to ensure that the end-user has no issues utilizing it. Many of the challenges that must be addressed or handled before attempting to devise a comprehensive design solution are described in this section.

6.2.1 Assumption and Dependencies

The assumptions and dependencies related to our mobile application that require attention are listed below.

6.2.1.1 Operating System

Users must have devices running Android 6.0 (Marshmallow) or later for Android, and iOS 15 or later for Apple devices, ensuring compatibility with the advanced features and optimizations incorporated into the app.

6.2.1.2 End User Characteristics

As the application is developed mainly for kids so it has few end-user considerations that are as follows:

- End users will have stable Internet Communication.
- End users should have a good device either Android or iOS that should support minimum requirements to install applications.
- End users should have a basic knowledge of using a mobile application

6.2.1.3 Possible Changes in Functionality

Potential functional modifications may occur while our mobile application is being developed as a result of changing technology, user input, and market needs. Maintaining alignment with new operating system upgrades, technology breakthroughs, and user expectations requires constant adaptation and development. Furthermore, as the user base grows, input from parents and guardians may influence changes to features and content delivery, improving the kids' overall educational and enjoyable experience. Maintaining the application's relevance and efficacy will need frequent updates and a responsive approach to user requirements.

6.2.2 General Constraints

FictionGenie aims to provide users with an easy-to-use and smooth platform to read stories and generate content with an interactive user experience. Nonetheless, there are some restrictions or limitations that will greatly affect how the system's software is designed. It is important to remember that any of the following could impose such restrictions.

6.2.2.1 Hardware and Software Environment

Following are some hardware and software limitations or constraints that our system might face and these could affect its performance.

- Availability of a stable Internet Connection.
- Device specifications diversity (RAM, storage, processor speed).

- Compatibility of applications with different Android and iOS versions.
- Adherence to different security and regulations.
- Optimization issues for devices with low processing power.
- Consideration for third-party service dependencies (text-to-speech, translation).

6.2.2.2 End-User Environment

Following are some limitations that could be at the end of our users and will be limited to them.

- The user does not have a good and stable Internet Connection.
- The user does not have a basic understanding of operating a mobile application.

6.2.2.3 Availability of Volatility of Resources

Following are some points that should be taken care of regarding the availability and volatility of resources for the system.

- Low bandwidth internet connection.
- Devices with slow processing power.
- Availability issues of servers and critical services.

6.2.2.4 Standards Compliance

Following are some points that should be taken care of regarding the compliance of our application with proper standards and guidelines.

- Application should be updated regularly according to the latest industry standards, frameworks, and protocols to stay intact with trends and security measures.
- Coding standards must be followed that are industry-recognized as code quality plays an important role in the life of an application.

6.2.2.5 Interoperability Requirements

Following are some Interoperability requirements for the system.

- There should be a smooth connection between the user and the application along with a smooth connection between the application and the third-party services that are being used by it to generate stories and provide other key functionalities.

- The application should observe cross-platform compatibility like it should function equally well on Android and iOS devices.
- Well-documented APIs should be used that allow easy integration with external services, enabling the app to leverage additional functionalities.

6.2.2.6 Interface/Protocol Environments

Following are some points regarding the Interface and protocol environment of the system.

- User Interface should be easy to navigate as most of the users will be kids.
- User Interface should be responsive so that users may enjoy a consistent and positive experience on tablets and mobile devices.

6.2.2.7 Data Repository and Distribution Requirements

Following are some considerations regarding the requirements of the system's database or repository.

- Implementation of a secure, efficient, and scalable database architecture must be done to handle growing volumes of user data, stories, preferences, and other relevant information.
- Utilization of a Content Delivery Network (CDN) must be brought into practice to decrease latency. It would also lead to rapid content delivery and provision of good user experience.
- The database must be kept secure to avoid data leakage and database attacks.
- Application of secure data transmission between the app and the server by implementing encryption protocols to ensure data integrity and privacy.

6.2.2.8 Security Requirements

Following are the security requirements associated with our application.

- User Authentication should be done by implementing a robust user authentication system to ensure secure access, requiring strong passwords.
- Encryption Algorithms should be implemented to protect sensitive user data, both in transit and at rest, as it would provide confidentiality and integrity.
- Implementation of strict access control mechanisms should be done to limit user privileges.

6.2.2.9 Memory and other capacity limitations

Following are the memory limitations associated with our application.

- The app may face limitations in terms of available memory space on devices as if the user does not have sufficient storage space in a mobile phone, potentially impacting performance and responsiveness.
- Users with limited storage capacity may encounter challenges in storing a large volume of stories, images, and other content within the application.

6.2.2.10 Performance Requirements

Following are some key requirements related to the performance of the application.

- The system should be able to handle loads of users.
- The system should handle improper user inputs accordingly.
- There must be no downtime.
- The system should implement and continuously optimize its algorithms to ensure a proper and safe connection between itself, third-party service providers, and the end users.
- The system should utilize efficient caching mechanisms to avoid redundant computations and enhance system performance.
- The system should be able to optimize resource utilization, including CPU, memory, and network usage, to ensure efficient operation and prevent performance bottlenecks.
- The system should implement offline capabilities for some features like reading pre-read stories or storing stories in case the user has no internet connection.

6.2.2.11 Network Communications

Following are some important points regarding network communications.

- Bandwidth limitations could affect the performance of the application as it will slow down the communications between the user and the app server.
- Strong error handling and offline capabilities are essential since the application's functioning may be affected in situations when the network connection is not up to the mark.
- Different latency issues could arise in the network communications which should be handled.

6.2.2.12 Verification and Validation Requirements

Following are some important points regarding the verification and validation requirements associated with our application.

- We should ensure comprehensive testing across different devices with different parameters like OS, and screen sizes to guarantee responsiveness.
- Proper security tests and vulnerability assessments must be done.
- User Acceptance Testing must be brought into practice to validate that the application meets user expectations and that user feedback could be used to improve the application which will increase its use-ability,

6.2.3 Goals and Guidelines

Following are some goals and guidelines that could dominate the design of the system's software.

6.2.3.1 The KISS principle ("Keep it simple stupid!")

The goal is to adhere to the KISS principle. It would be done to make our application design as simple as possible so the software could be maintainable and less complex.

6.2.3.2 Scalability

The goal is to make our system a highly scalable one so a lot of users are able to use our application at the same time without any compromise on its performance or availability.

6.2.3.3 User-Centric Design

We should give utmost priority to the experience of users so we have to develop a very engaging and easy-to-navigate user interface. As the target users are kids a complex user interface will not be beneficial for them.

6.2.4 Development Methods

The project's development will follow the Scrum approach, an agile methodology that prioritizes flexibility and iterative progress. This methodology is particularly beneficial for software development, relying on repeatable cycles known as sprints. Each sprint will have a defined deadline, ensuring that work progresses in manageable increments. Notably, these sprints will be conducted within a week, facilitating a rapid and adaptive development process. Post each sprint, a concise 15–20-minute meeting will convene to assess progress and make any necessary adjustments to the overall project schedule. Additionally, daily sprint meetings will be conducted, providing a forum to discuss specific objectives for the day, fostering collaboration and keeping the team aligned with project goals. The use of Scrum methodology positions the project for adaptability, enabling adjustments to system requirements at any

stage of development to accommodate evolving needs and ensure the successful realization of project objectives.

6.3 System Architecture

The system is being made to provide a platform where users can read stories and generate the stories or tailor them according to their own choices. The user experience is enhanced in the system by the introduction of key components like text-to-speech, multilingual support, and parental monitoring. User has the option to read stories that are pre-built in the application or the user can generate stories as per their choice. The prebuilt stories are stored in the cloud database of the application whereas the AI-generated stories are fetched via a Large Language Model at run time and will be stored in the database along with the ID of the user who has generated it based on the prompts. Features like text-to-speech and multilingualism will operate once the user has opted to choose them, these are mainly provided by a third party. The detailed system architecture diagram is provided below:

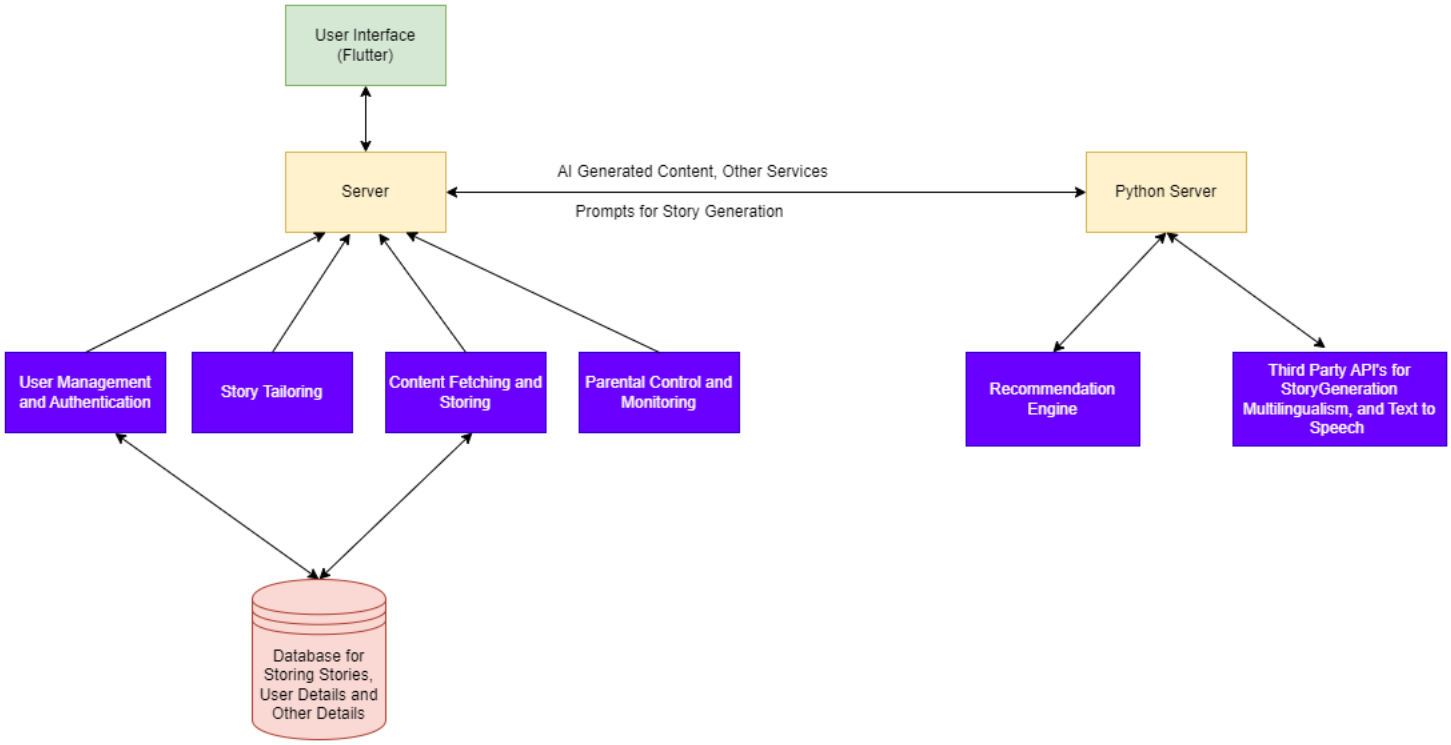


Figure 6.1: System Architecture Diagram

This figure shows the System Architecture Diagram of Application

6.3.1 User Interface

This module includes UI components developed using the Flutter framework for a cross-platform mobile app.

6.3.2 Server

This server manages sub-systems like user handling, real-time database functionality, and cloud functions for serverless backend logic.

6.3.3 User Management and Authentication

This is a separate component for managing user profiles, authentication, and user-related functionalities. It queries data from the database and verifies user identities who are interacting with the application and also stores new users in the database.

6.3.4 Story Tailoring

A component for handling stories that are generated based on user prompts and tailoring stories according to user preferences. It also tailors stories in a way that they have proper synchronization with the format of stories that are to be followed.

6.3.5 Content Fetching and Storage

This component retrieves prebuilt and AI-generated stories from the database. It also stores user-generated stories in the database.

6.3.6 Parental Control and Monitoring

It is the component dedicated to implementing parental control features and monitoring tools.

6.3.7 Python Server

It serves as the central hub for coordinating various business logic components and handling API requests and other requests like story generation from the other server.

6.3.8 Recommendation Engine

This component generates personalized story recommendations for users. It also utilizes user profiles, preferences, and historical data to suggest relevant content.

6.3.9 Third-Party Services

Third-party services will be an important component of the system which includes Gemini which will generate stories based on the inputs provided to it. It also includes AI to generate images on the basis of story content using APIs. It also includes two services: multilingualism and text-to-speech will will be provided by Google API.

6.4 Architectural Strategies

After thorough research following strategies are decided for the architecture of our system.

6.4.1 Programming Languages

Two main programming languages will be used for the development of the application that are Python and Dart with Flutter. The use of Flutter guarantees a unified user experience on the iOS and Android operating systems. The hot-reload function of Flutter speeds up development, enabling quicker iterations and more effective debugging. For the backend services, using Python—especially with the Flask framework—offers a scalable and lightweight solution. Flask's ease of use and rapid development make it an excellent choice for developing RESTful APIs and implementing business logic.

6.4.2 Cloud-Based Database

Google Firebase will be used as the cloud-based database to provide scalability, user-friendliness, and real-time data synchronization. It offers a NoSQL database for storing user profiles, stories, AI-generated images, and other application data, and it connects with Flutter with ease.

6.4.3 Error detection and Recovery

Comprehensive logging and monitoring will be done with regard to error detection and recovery. The aim will be to provide fast diagnosis and enable effective recovery procedures when problems occur by collecting comprehensive information about mistakes, exceptions, and system behaviors in real time.

6.4.4 AI Integration with Large Language Models (Gemini)

We can generate powerful and engaging stories using large language models, such as Gemini.With this AI-driven method, stories may be created dynamically in response to user input also it generates user-centric stories and we can also give it feedback so stories our according to the requirements.

6.4.5 Scalability and Load Balancing

The application should be scalable so it can handle increased user loads. Load balancing techniques, such as those provided by Firebase or AWS, can be employed to manage traffic efficiently.

6.4.6 Concurrency and Load Balancing

We can use safe coding techniques, and synchronization mechanisms to provide concurrency. Moreover, we can use an efficient load-balancing strategy like round robin to manage traffic across servers.

6.5 Domain Model/Class Diagram

This section illustrates the class diagram of our system. It depicts different types of relationships like association, composition, and aggregation among different classes. The figure is shown below

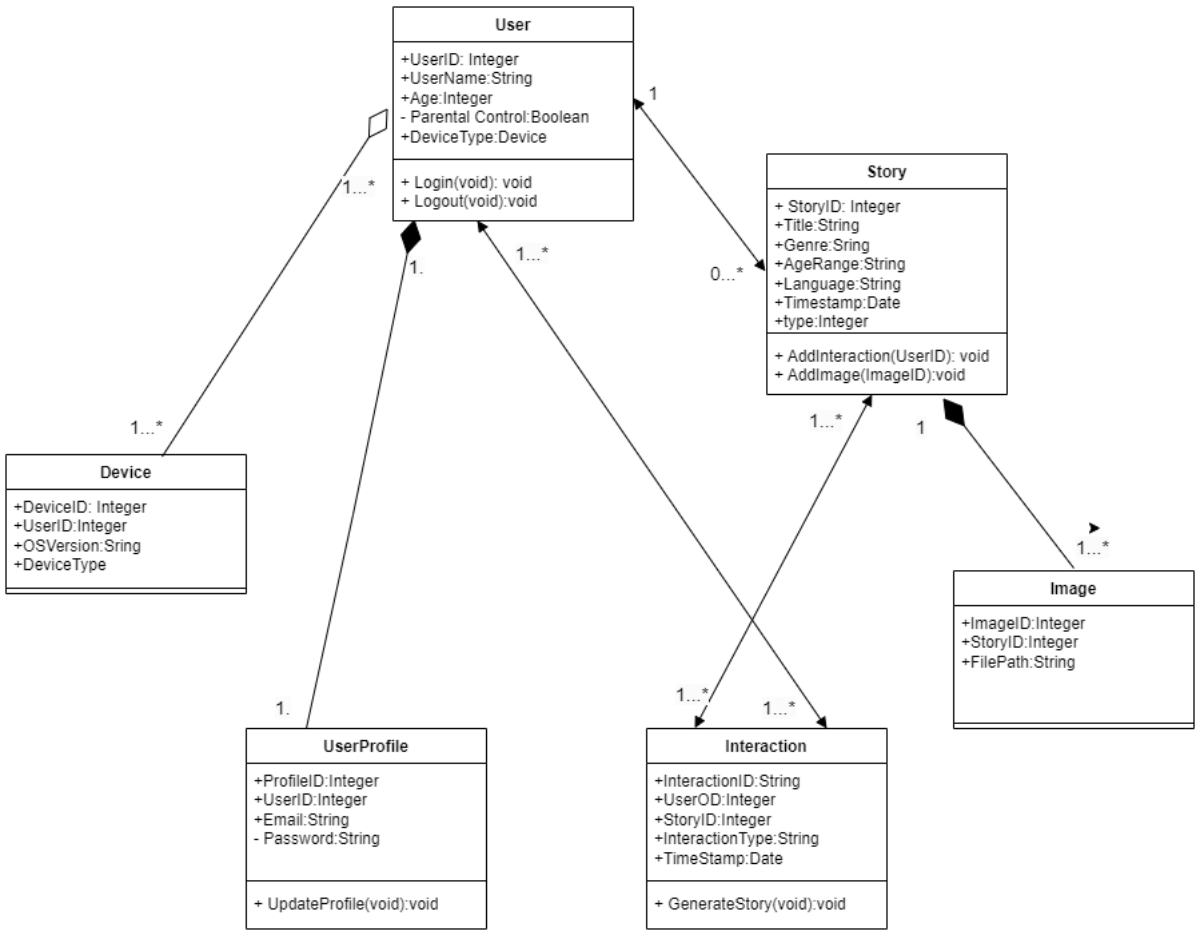


Figure 6.2: Class Diagram

This figure shows the Class Diagram of Application

6.6 Sequence Diagram

This section illustrates the sequence diagrams related to the different use cases of our system.

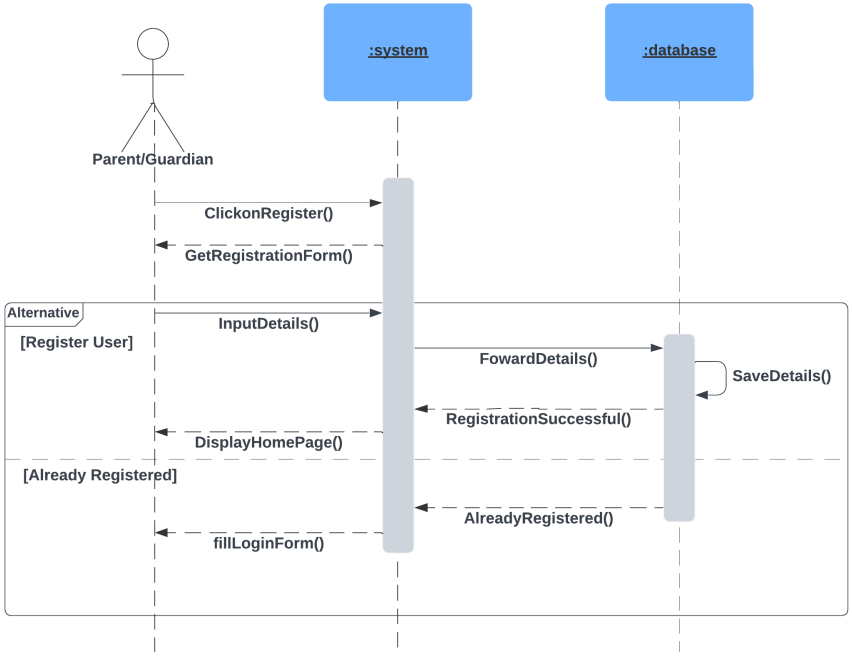


Figure 6.3: User Registration

This Sequence Diagram explains the whole process for registering a new user in the application.

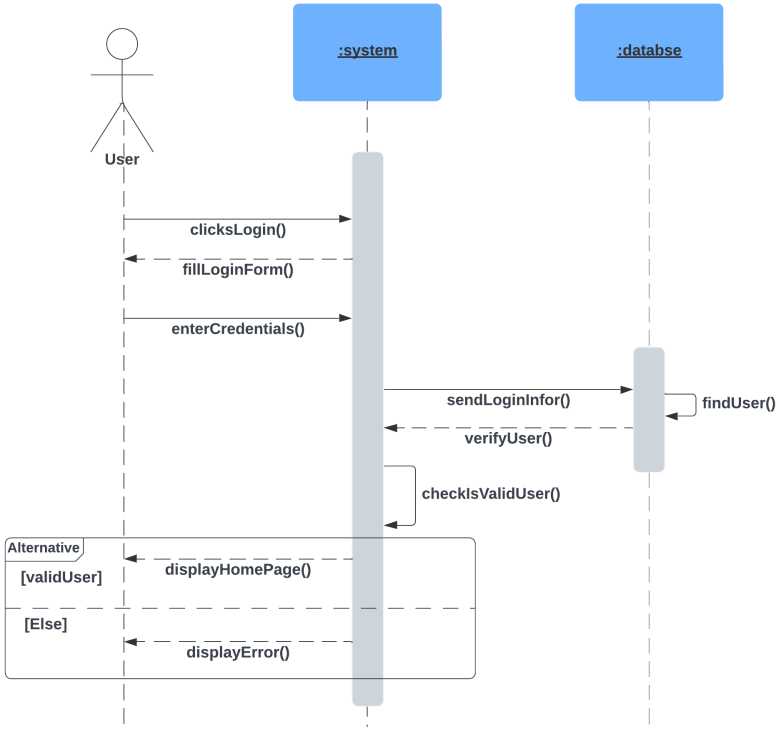


Figure 6.4: User Login

This Sequence Diagram explains the whole process for registering a new user in the application.

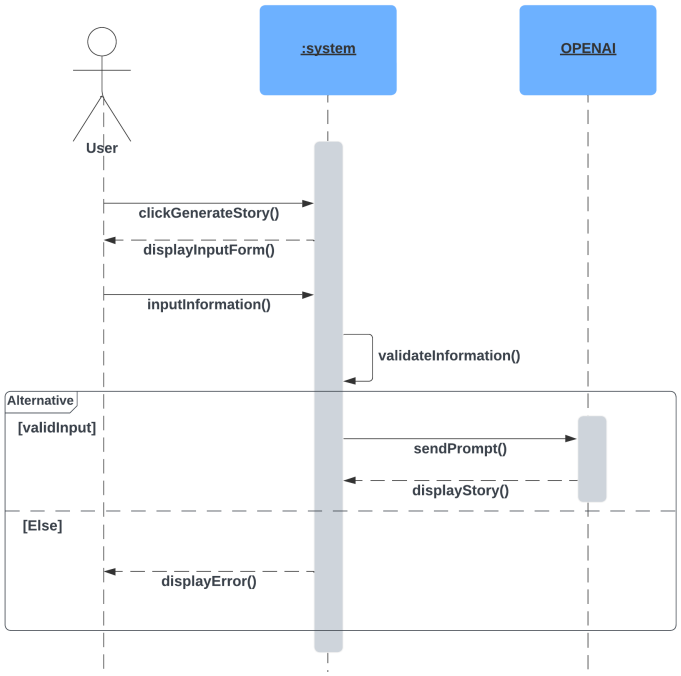


Figure 6.5: Story Generation

This Sequence Diagram explains how a story is generated after user input.

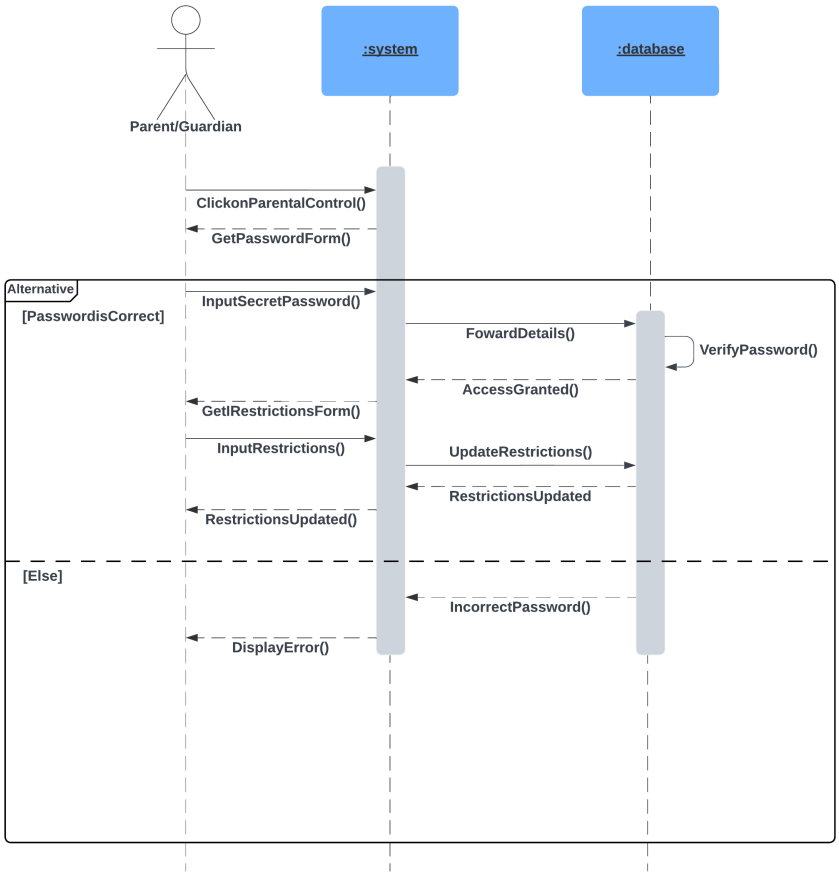


Figure 6.6: Parental Control

This Sequence Diagram explains how parents will be able to restrict their children's activity.

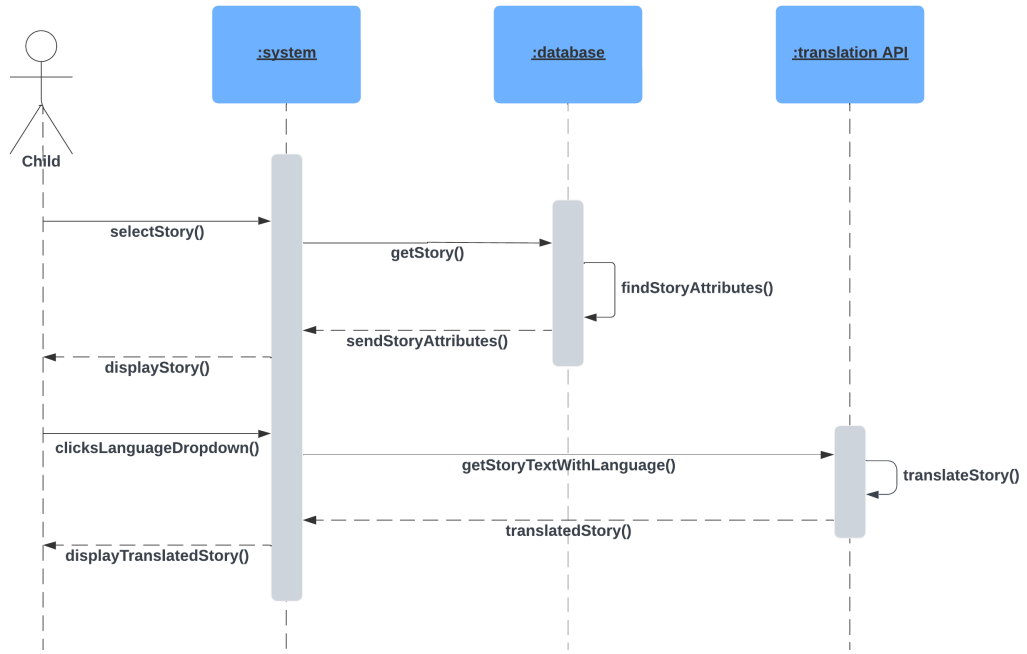


Figure 6.7: Multilingual Story Selection

This Sequence Diagram explains how children will be able to read stories in different languages from the choice provided.

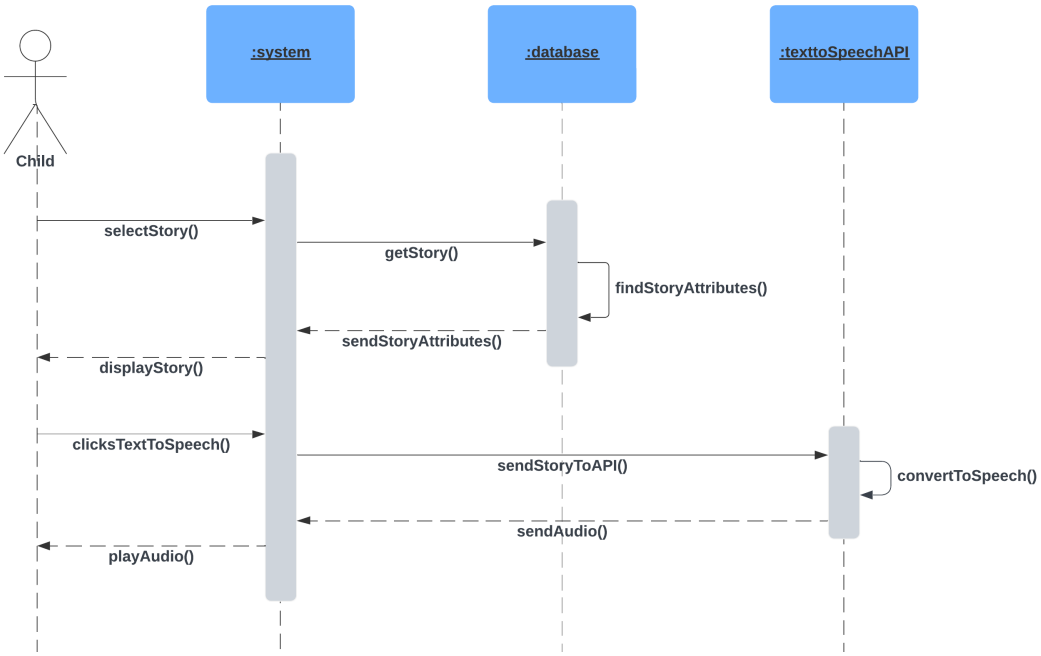


Figure 6.8: Text to Speech

This Sequence Diagram explains how children will be able to listen to stories.

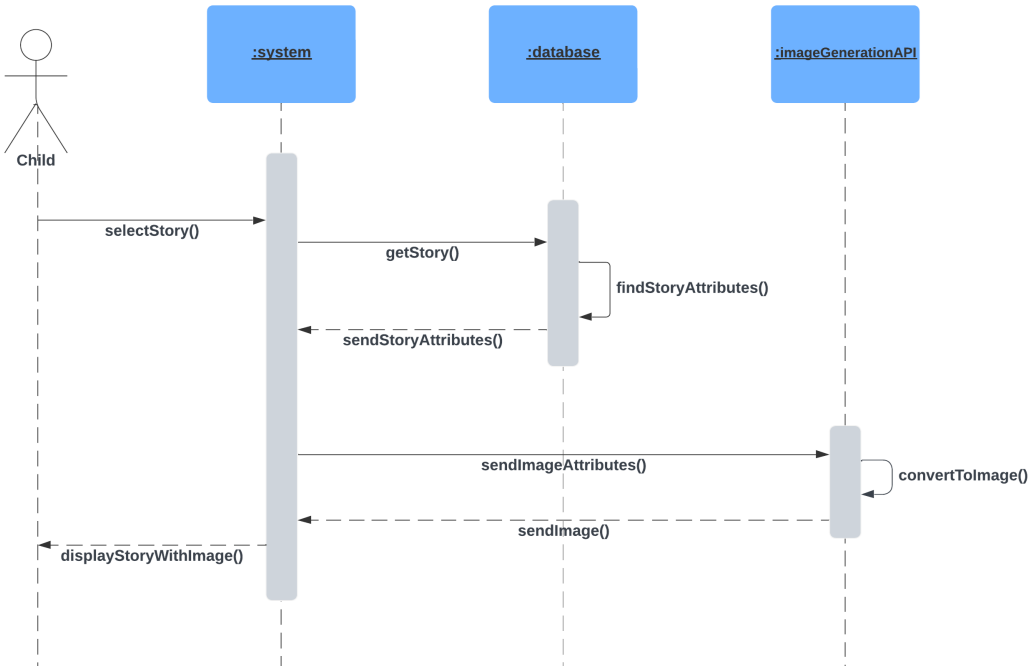


Figure 6.9: Dynamic Image Generation

This Sequence Diagram explains how images in background will be generated dynamically after story creation.

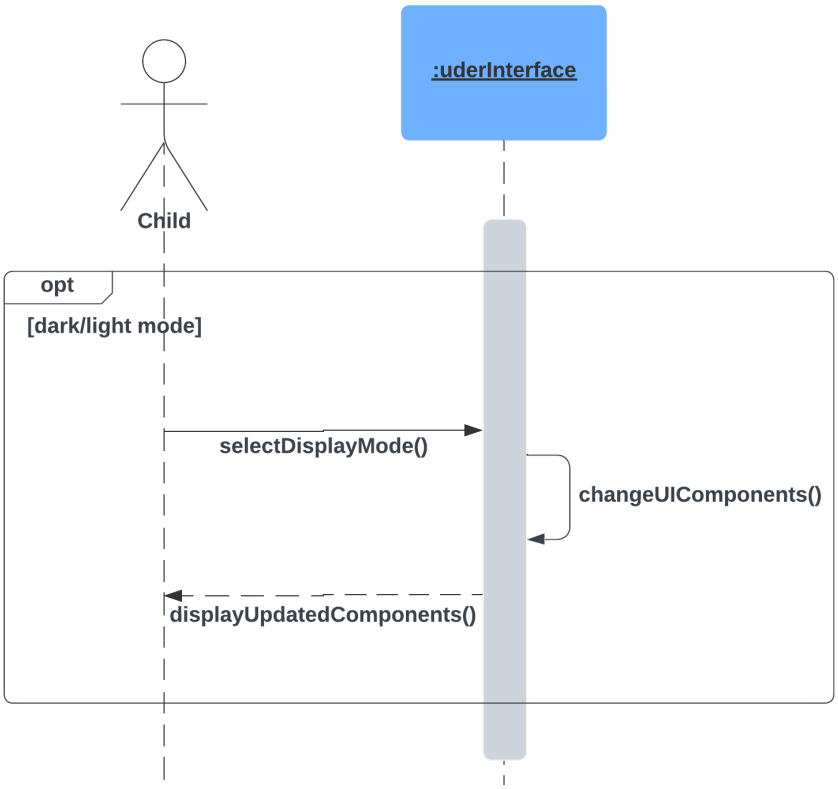


Figure 6.10: App Interface Dark/Light Mode

This Sequence Diagram explains how user will be able to change the theme of the application.

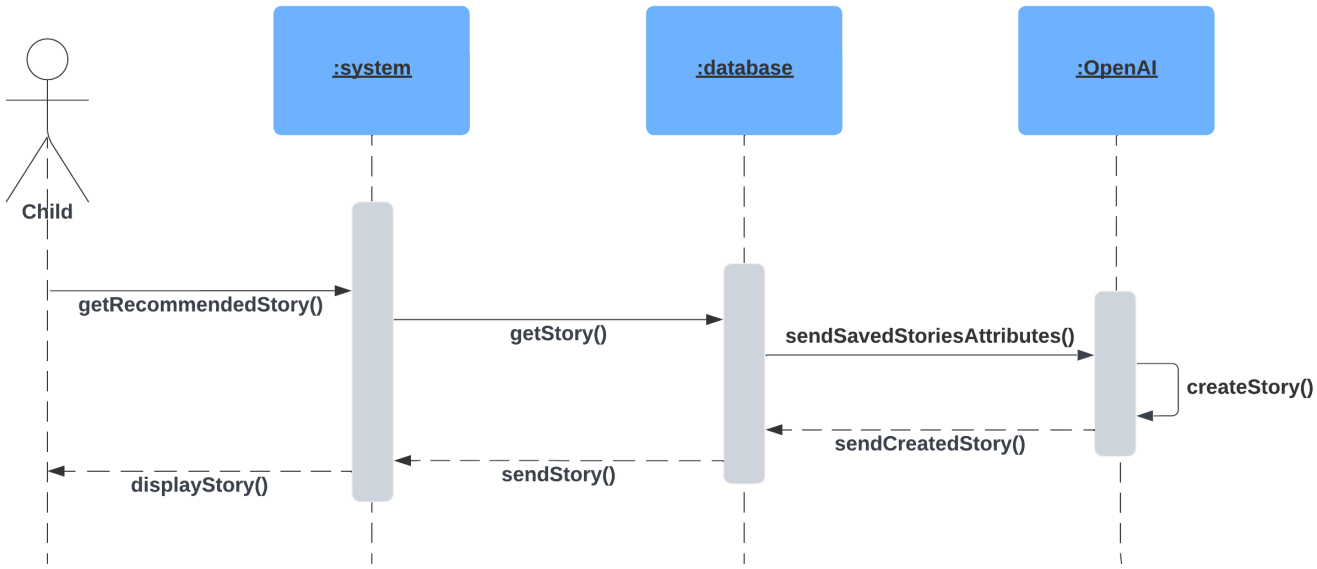


Figure 6.11: Story Recommendation

This Sequence Diagram explains how stories will be recommended to the user based on previously read stories.

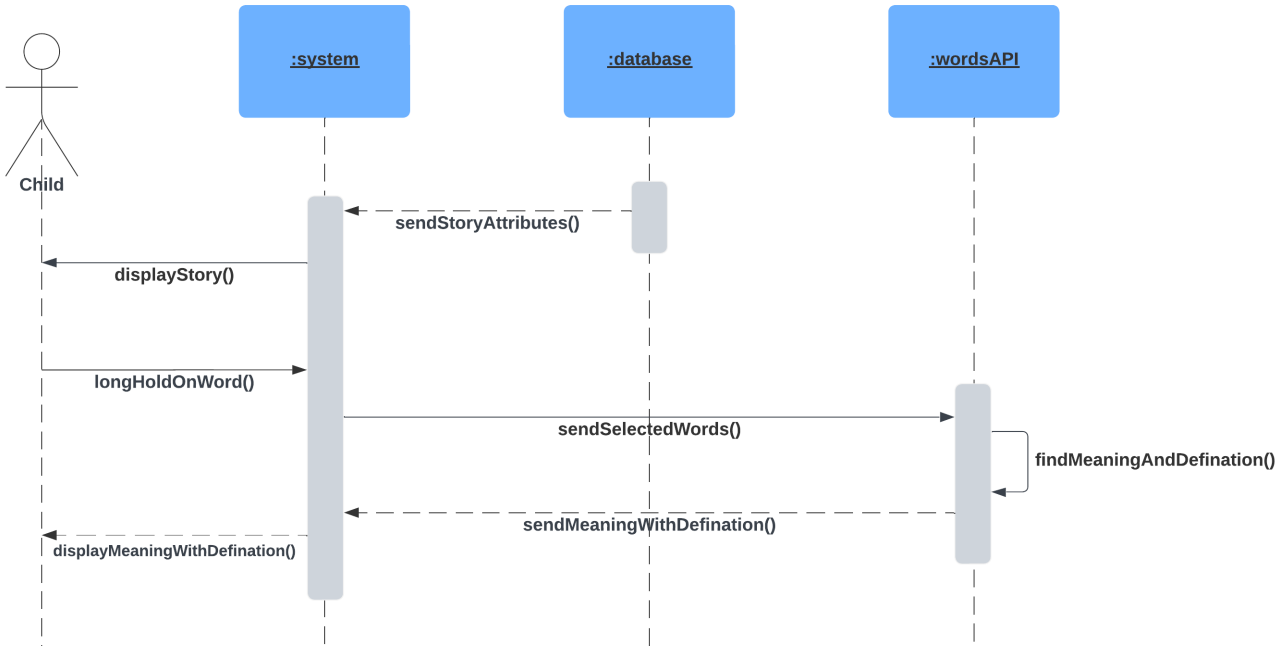


Figure 6.12: Instant Word Definition

This Sequence Diagram explains how children will be able to read definitions of words.

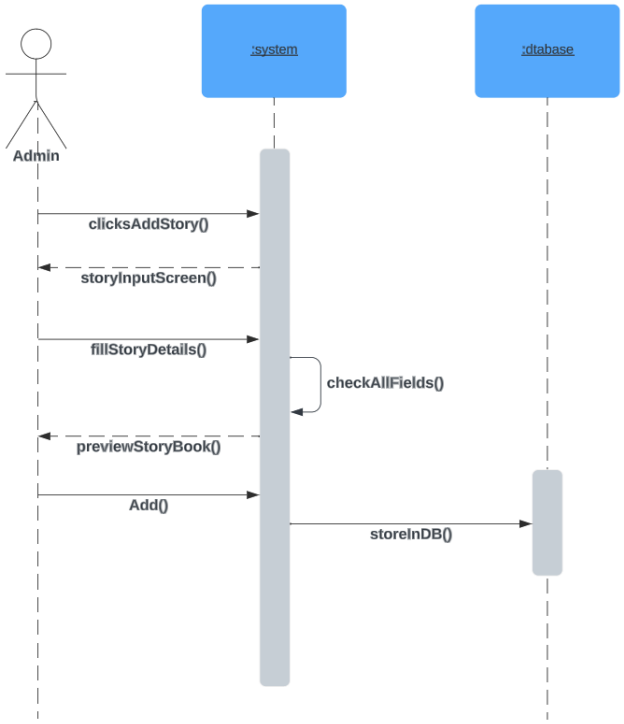


Figure 6.13: Add New Stories in the Application

This Sequence Diagram explains how the admin will add new story books into the application.

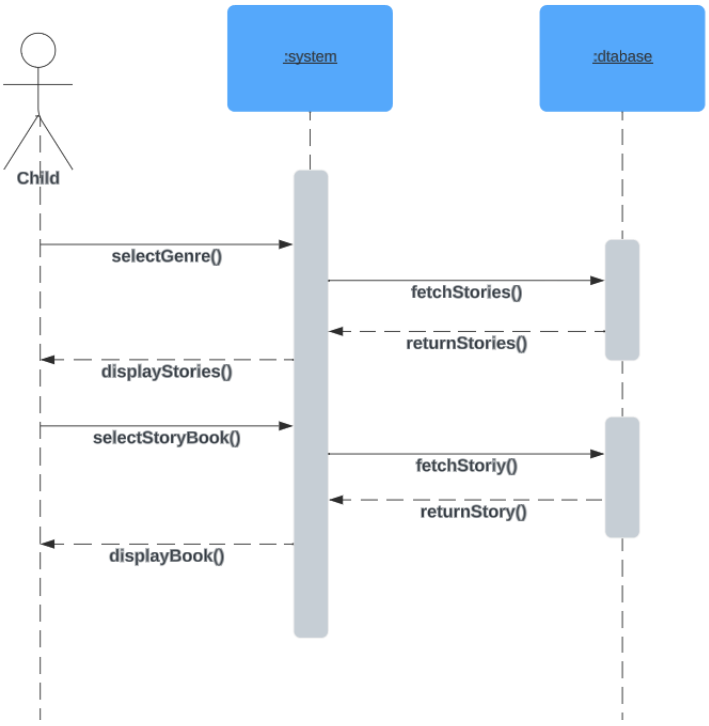


Figure 6.14: Read Pre-built Stories

This Sequence Diagram explains how children will be able to read the pre-built story books from the application.

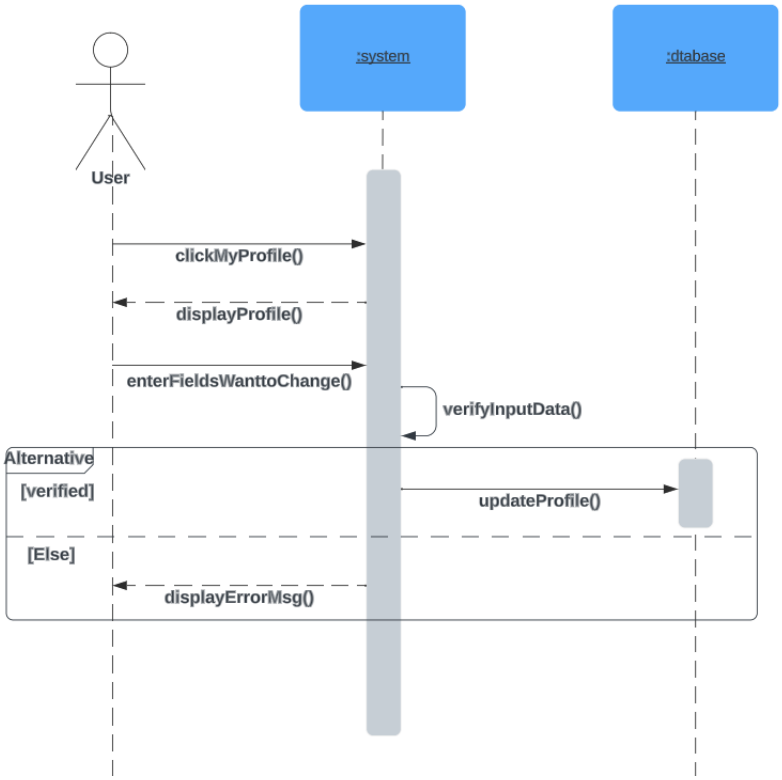


Figure 6.15: Update Profile

This Sequence Diagram explains how the users will be able to update their profiles, reset passwords and other details.

6.7 Policies and Tactics

The following factors would be taken into account for the project's policies:

6.7.1 Tools to be Used

During the development of FictionGenie many tools, APIs, and libraries will be brought into use. The most significant ones are as follows:

- OpenAI's DAL, a powerful image generation model, will be employed for creating visually engaging and unique images based on user prompts.
- Google Cloud Text-to-Speech API will be integrated for the smooth narration of stories.
- Google Cloud Translation API will be utilized for multilingual support, enabling users to enjoy FictionGenie in their preferred language.
- Google Gemini, a Large Language Model, will dynamically generate stories based on user inputs, fostering interactive and personalized storytelling.

- Python's TensorFlow for recommendation engine will be used This engine will analyze user preferences and historical data to suggest relevant and captivating stories, enhancing the overall user experience.

6.7.2 Coding guidelines and conventions

Python and Flutter will be the two programming languages used. Safe coding practices shall be strictly adhered to together with standard set and programming regulations.

6.7.3 User Interface

The user interface of the application will be easy to navigate and visually appealing to the users. As the application is mainly for young users bright colors will be used on most of the screens or UI components.

6.7.4 Extensibility

The architecture will be extensively built to allow for the easy integration of new features and modules, guaranteeing that the application may readily change and adapt to requirements and enhancements in the future.

6.7.5 Policies for System Testing

A comprehensive system testing policy will be developed for the FictionGenie application, involving unit, integration, and end-to-end tests, with comprehensive documentation and automated procedures for uniformity and effectiveness.

6.7.6 Software maintenance

A comprehensive set of policies will be implemented to guarantee the continuous well-being and functionality of the FictionGenie application in the domain of software maintenance. Regular updates, app fixes, and different schemes of optimization will be used to resolve vulnerabilities in software, improvement of security mechanisms, and provision of new features. Version control protocols will be used in efficient bug tracking to enhance the environment for software maintenance over time.

6.7.7 Algorithms

We will use the efficient algorithms in our system that will be efficient in terms of time and space complexity. For searching stories in the huge database of stories we will use efficient pattern matching algorithms and searching algorithms like binary search. Big-O analysis will be done, to evaluate the

scalability and efficiency of the algorithm we have chosen. Our application will continue to improve its algorithms with time to improve its scalability and performance. Collaborative filtering will be initially used regarding the story recommendation to the users based on their preferences for stories. The major focus will be on the implementation of efficient algorithms in the high-end features of our application like recommendation system, narrative creation, and image processing.

6.8 Conclusion

In conclusion, this chapter has thoroughly discussed the system architecture, high and low-level design, goals, implementation techniques, tools, and policies related to the development, and testing, maintenance of the system. High-level design emphasizes user-centric features like story creation, AI model, and API integration, while low-level design focuses on details like data structures, algorithms, designing different classes, and coding techniques. These design stages provide a stable, extendable architecture for our application, enabling effective AI-driven storytelling. The knowledge regarding the combination of high-level and low-level design approaches in the system design ensures the system's adaptability to the requirement changes and user needs.

Chapter 7 Implementation and Test Cases

This chapter discusses the work that have been done by the team so far along with some technical details and back-end modules that were not discussed in the high and low-level design. The content in this chapter is not final and can evolve as we go on with implementing more modules.

7.1 Implementation

During the project's implementation phase, our team has put a dedicated effort in developing and exploring different tools and utilities that will help us in the further implementation of our application. The team investigated several technologies and APIs to construct and improve the functionality of FictionGenie. The implementation details of the work done so far are below:

7.1.1 Pre-built Stories Library

The team implemented and tested a Pre-built Stories Library module. This module sought to increase user engagement by presenting a selection of 30 classic children's stories divided into five major genres: mystery, fantasy, fairy tale, science fiction, and historical fiction. To assess system performance and user engagement, 10 users interacted with the program, comprising 6 children aged 8 to 10 and 4 adolescents aged 11 to 15. Users were encouraged to read stories that matched their genre interests. User interactions, engagement levels, preferences, and the overall impact of this module on the application were all examined.

7.1.2 Exploring AWS Lambda

The team explored the use of AWS Lambda for the initial component of our system, which involves the development of AI-based stories depending on user inputs. We made a sample sample and ran it on AWS lambda to get started with the proper implementation of this component.

7.1.3 AI Based Stories

FictionGenie's team worked on the implementation of AI-based story generation using the power of Google's LLM i.e. Gemini. The team has made major progress in the development due to this module. We take different inputs from users like the title of the story, genre(s), difficulty level, no of characters, and other keywords. The inputs are given to the AI model after the application of different prompt engineering techniques to generate a relevant, concise, and suitable story for the children. Moreover, we have also incorporated some methodologies to make sure that the stories generated by AI give a feel that

an actual author has written them.

7.1.4 AI Generated Images

FictionGenie's team has put in a lot of effort to incorporate the AI-generated images module in the context of story generation. We have integrated the Open AI's model DALLE-3 for image generation based on the content of stories. The images generated are relevant to the scenarios of the story and these images ultimately enhance the reading experience of a child and attract him/her more towards our application.

7.1.5 User Registration

We have implemented the user registration module of our application. A user is displayed a signup page where a user is prompted to enter some relevant details and based on this after processing at the backend a user is signed up once all the checks are completed. Once the user is signed up he also receives a confirmation email. The details of the user are stored in Google Firebase.

7.1.6 User Authentication

We have implemented the user authentication using the Google Firebase. We have crafted a login page based on the user's email and password it authenticates the user allowing him to go toward the home page of our application and enjoy the experience.

7.1.7 Profile Customization

We have implemented a very interesting feature that allows the user to pick an avatar as the profile icon for the customer the profile and also select a preferred theme for the stories. Moreover, users have the option to edit any relevant detail of their profile at any time.

7.1.8 Favorite Stories

We have implemented a feature known as add to favorites where a user can mark a story as favorite and that story is kept in the application so whenever a user goes to his favorites section all the stories that were marked by him as favorites are shown to him without the need of remembering or searching them again.

7.2 Test Case Design and Description

All of the test cases performed are on Android mobile phones equipped with a minimum of 6GB RAM and a high-performance processor, ensuring optimal conditions for testing. The results obtained from these tests serve as a benchmark for devices meeting or exceeding these specifications. However, it's important to note that performance may vary on devices with different hardware configurations, particularly in terms of processing power and speed.

As the application is currently in its development phase and has not been published on any app distribution platform, the APK file of the application is manually installed on each test device. This allows for thorough testing under controlled conditions, enabling us to identify and address any potential issues before release.

We have conducted extensive testing procedures to validate the functionality of every implemented feature and to detect and rectify any bugs or anomalies. Our testing process encompasses a wide range of scenarios and use cases to ensure that the application meets the highest standards of quality and reliability. The test cases performed on our application are enlisted below:

Table 7.1: Setting-Up Child Profile

This test outlines cases for creating and managing a child profile in the application.

User Registration			
Test Case ID:	*1*	**QA Test Engineer:**	*Zaid Khan*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*20/04/2024*	**Use Case Reference(s):**	*User Registration*
Revision History:	*None*		
Objective:	*To test if the user is able to register and setup child profiles.*		
Product/Ver/ Module:	*FictionGenie/1.0/RegistrationModule*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*None*		
Step No.	**Execution description**	**Procedure result**	
1	*The user enters valid name, email, password and PIN.*	*The user is sent a verification email.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.2: Restrict Contents for Children

This test outlines cases for functionalities related to content restriction for child profiles

Story Generation			
Test Case ID:	*2*	**QA Test Engineer:**	*Muhammad Usama*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*20/04/2024*	**Use Case Reference(s):**	*Story Generation*
Revision History:	*None*		
Objective:	*To test if explicit content is visible to children.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryGenerationModule*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*Child is logged into the profile.*		
Step No.	**Execution description**	**Procedure result**	
1	*The user sends a prompt to generate story.*	*User will be asked to generate story again.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.3: Child Activity Dashboard for Parents

This test outlines cases related to dashboards for parents.

Parental Control			
Test Case ID:	*3*	**QA Test Engineer:**	*Muhammad Usama*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*20/04/2024*	**Use Case Reference(s):**	*Parental Control*
Revision History:	*None*		
Objective:	*To test if child's analytics are visible to parents.*		
Product/Ver/ Module:	*FictionGenie/1.0/UserProfile*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User has entered correct pin.*		
Step No.	**Execution description**	**Procedure result**	
1	*The user is on Profile view page.*	*User will be able to view child analytics.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.4: Forget Password

This test outlines cases in the case when a user forgots the password while logging in.

Login			
Test Case ID:	*4*	**QA Test Engineer:**	*Zaid Khan*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*21/04/2024*	**Use Case Reference(s):**	*Login*
Revision History:	*None*		
Objective:	*To test if the user is able to login after he/she forgot the password.*		
Product/Ver/ Module:	*FictionGenie/1.0/UserProfile*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is registered on the application.*		
Step No.	**Execution description**	**Procedure result**	
1	*User is directed on forgot password page.*	*User gets password update email.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.5: Select Avatar

This test outlines cases for selecting a avatar for a child profile.

User Registration			
Test Case ID:	*5*	**QA Test Engineer:**	*Zaid Khan*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*22/04/2024*	**Use Case Reference(s):**	*User Registration*
Revision History:	*None*		
Objective:	*To test if an avatar is associated to child profile.*		
Product/Ver/ Module:	*FictionGenie/1.0/UserRegistration.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User has already provided registration detail.*		
Step No.	**Execution description**	**Procedure result**	
1	*User has selected avatar and moves to homepage.*	*In Profile Avatar will be updated*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.6: Read Stories

This test outlines cases for reading a story book within the application.

Read Stories			
Test Case ID:	*6*	**QA Test Engineer:**	*Zaid Khan*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*22/04/2024*	**Use Case Reference(s):**	*Read Stories*
Revision History:	*None*		
Objective:	*To test if user is able to read story books.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryBookModule.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
1	*User has selected a story book or generated a story.*	*Story is visible to User*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.7: Listen to a Story

This test outlines cases and functionalities provided to user to listen to a story book in audio form.

Text-to-Speech			
Test Case ID:	*7*	**QA Test Engineer:**	*Zaid Khan*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*22/04/2024*	**Use Case Reference(s):**	*Text-to-Speech*
Revision History:	*None*		
Objective:	*To test if the user is able to listen to stories in audio form.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryBookModule.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is on story page.*		
Step No.	**Execution description**	**Procedure result**	
1	*User has pressed text to speech icon*	*User will be able to listen to stories.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.8: Translate a Story

This test outlines cases and all the functionalities related to translating a story book in multiple languages.

Multilingual Stories Selection			
Test Case ID:	*8*	**QA Test Engineer:**	*Muhammad Usama*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*23/04/2024*	**Use Case Reference(s):**	*Multilingual Stories*
Revision History:	*None*		
Objective:	*To test user is able to translate stories into multiple languages.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryBookModule.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is on story page.*		
Step No.	**Execution description**	**Procedure result**	
1	*User has pressed multilingual language icon.*	*Text is converted into selected language.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.9: Generate Story

This test outlines cases for generating a story by a child with in the application.

Story Generation			
Test Case ID:	*9*	**QA Test Engineer:**	*Muhammad Usama*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*23/04/2024*	**Use Case Reference(s):**	*Story Generation*
Revision History:	*None*		
Objective:	*To test if the user is able to generate own stories.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryGenerationModule.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is on story page.*		
Step No.	**Execution description**	**Procedure result**	
1	*User has sent a prompt to generate story.*	*Story is displayed to User.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.10: Story Recommendation

This test outlines cases and all the functionalities related to recommending stories to a user.

Story Generation			
Test Case ID:	*10*	**QA Test Engineer:**	*Zaid Khan*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*24/04/2024*	**Use Case Reference(s):**	*Story Generation*
Revision History:	*None*		
Objective:	*To test if the user is able to see recommended stories.*		
Product/Ver/ Module:	*FictionGenie/1.0/Homepage.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is logged-in.*		
Step No.	**Execution description**	**Procedure result**	
1	*User is on home page*	*Recommended Stories will be displayed.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.11: Image Generation

This test outlines cases and all the functionalities related to Image generation.

Story Generation			
Test Case ID:	*11*	**QA Test Engineer:**	*Muhammad Usama*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*24/04/2024*	**Use Case Reference(s):**	*Story Generation*
Revision History:	*None*		
Objective:	*To test if the user is able to see dynamic image of story.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryGenerationMdule.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is has generated a story..*		
Step No.	**Execution description**	**Procedure result**	
1	*User is on generate story page*	*Image related to story displayed.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.12: Word Definition

This test outlines cases and all the functionalities related to instant word definition and meanings.

Story Reading			
Test Case ID:	*12*	**QA Test Engineer:**	*Muhammad Usama*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*25/04/2024*	**Use Case Reference(s):**	*Story Reading*
Revision History:	*None*		
Objective:	*To test if the user if able to get word definition and meanings.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryReadingModule.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is logged-in.*		
Step No.	**Execution description**	**Procedure result**	
1	*User has selected a word on story page*	*User is able to see meaning of word.*	
Comments: The system worked properly according to our requirements.			
Passed			

Table 7.13: Dark/Light Mode

This test outlines cases and all the functionalities related to application dark/light mode.

Dark/Light Mode			
Test Case ID:	*13*	**QA Test Engineer:**	*Muhammad Usama*
Test case Version:	*1.0*	**Reviewed By:**	*Usman Azeem*
Test Date:	*25/04/2024*	**Use Case Reference(s):**	*Story Reading*
Revision History:	*None*		
Objective:	*To test if the user if able to switch modes light/dark.*		
Product/Ver/ Module:	*FictionGenie/1.0/StoryReadingModule.*		
Environment:	*Android 12 Operating System.*		
Assumptions:	*None*		
Pre-Requisite:	*User is logged-in.*		
Step No.	**Execution description**	**Procedure result**	
1	*User has selected light/dark mode toggle button*	*User is able to see background color change*	
Comments: The system worked properly according to our requirements.			
Passed			

7.3 Test Metrics

Following test matrices shows the summary of test cases and their results.

Table 7.14: Functional Test Case Metrics

This table shows the summary of functional test cases.

Metric	Purpose
Number of Test Cases	12
Number of Test Cases Passed	12
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	0

Table 7.15: Non-Functional Test Case Metrics

This metrics shows the summary of non-functional test cases.

Metric	Purpose
Number of Test Cases	1
Number of Test Cases Passed	1
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	0

7.4 Conclusion

In conclusion, this chapter has thoroughly discussed about the implementations of FictionGenie performed. The AI based stories are generated along with dynamic images to captivate young minds. Further proper authentication system is used to ensure user details are kept safe. Further, in this chapter we have tested all of our functional and non-functional requirements. The tests are performed and also reviewed again to ensure a seamless user experience when using our application. We are glad to inform that all of the tests have passed and none of them failed. The test metric are also used to display the testing results.

Chapter 8 User Manual

This chapter will discuss the user manual of our application FictionGenie. This user manual will help users in operating our application and will aid them in case of any confusion.

8.1 Getting Started

To get started with FictionGenie on your Android or iOS device, please follow the below-mentioned instructions:

1. Download and Install the application from Google Play Store (for Android) and App Store (for iOS)
2. Once downloaded, launch the app by tapping the icon of the application.
3. Login or signup to access the features of FictionGenie.

8.2 Login

To Log in into the application, please follow these steps:

1. Locate the login button positioned at the center of the welcome screen and click on it to proceed to the login page.
2. Fill in the required fields, namely email and password, with your respective credentials. Once completed, click on the login button, usually labeled ”Login” or similar, positioned on the right side of the screen.
3. Optionally, consider checking the box labeled ”Remember Me” below the login fields. This will allow the application to remember your login credentials for future sessions, eliminating the need to enter them each time you access the application.

8.3 Signup

To create an account on FictionGenie, please follow these steps:

1. Locate the signup button positioned on the welcome screen. You can find it below the login button in the center of the screen or alternatively in the bottom left corner of the login screen. Click on it to proceed to the signup page for the creation of an account.
2. Fill in the required fields, including your name, email, password, and a 4-digit pin that will be

used to access some high-level features of the application, with your respective credentials. Once completed, click on the submit button, usually positioned on the right side of the screen. This action will navigate you to the Avatar selection screen.

3. Select an avatar as per your choice, it will be your profile icon.
4. After selecting the avatar, you will be redirected to the login screen. Proceed to login using the credentials you just created.

8.4 Forgot Password

In case you forgot your password, you can access your account by following these steps:

1. On the login screen, locate the ”Forgot Password” button positioned at the bottom right corner. Click on it to proceed to the ”Forgot Password” screen.
2. On the ”Forgot Password” screen, enter your email address in the designated field. This email address should be associated with your account.
3. After entering your email address, click on the ”Send Code” button. A verification code will be sent to the provided email address.
4. Check your email inbox for the verification code. Once received, enter the code into the corresponding field on the ”Forgot Password” screen.
5. After entering the verification code, you will be prompted to set a new password for your account. Enter your new password in the ”New Password” field and confirm it by re-entering it in the ”Confirm New Password” field.
6. Once you have entered and confirmed your new password, click on the ”Submit” button.
7. You will be redirected back to the login screen. Proceed to login using your updated credentials, consisting of your email address and the newly set password.

8.5 Reading a Pre-Built Story

To read stories, explore stories, and get in touch with features like dark mode, text-to-speech, and translation, please follow these steps:

1. After logging in, access a variety of stories on the home page, categorized under labels like ’Trending’, ’New’, and ’Popular’. Click on any story title to begin reading.
2. You can also use the search icon below to search for specific stories based on author name, title,

genre, or other filters, enhancing your browsing experience.

3. Once you select a story, you'll be directed to the 'Read Story' page where you can read the story.
4. Switch to dark mode for low-light reading by clicking on the dark mode button on the screen.
5. Translate the story into your preferred language by selecting the relevant language option at the top of the screen.
6. Enable text-to-speech functionality by clicking on the speaker icon at the top of the screen, allowing you to listen to the story instead of reading it.
7. Once done with reading, you can press the back arrow to go back to the home page of the application.

8.6 Generating an AI-based Story

To generate an AI-powered story , please follow these steps:

1. To generate an AI-based story, locate the "+" icon at the middle bottom of the home page or access the menu bar of the application. Clicking on either will navigate you to the AI-based story generation page.
2. On this page, you'll find a form where you can input the topic of the story, the desired number of characters, select genres, specify keywords for the story, and choose the difficulty level from the dropdown options (easy, medium, hard).
3. After filling in the required details, press the submit button to generate an AI-based story.
4. Once the story is generated, you will be redirected to the reading page where you can access all the features described in the above section, such as dark mode, language translation, and text-to-speech functionality.
5. You can also give a rating to the generated story based on your satisfaction level.

8.7 Add to Favourites

To add a story to your favourites, please follow these steps:

1. While reading a story, locate the "Add to Favorites" button, typically represented by a "heart" icon, on the story page.
2. Click on this button to mark the story as a favorite.

3. Once added to your favorites, the story will be easily accessible from the favorites section of your account or you can access the favorites sections from the menu bar.
4. You can remove a story from your favorites by clicking on the ”Remove from Favorites” option, usually available next to the story title.

8.8 Parental Monitoring

To access the Parental Monitoring feature of FictionGenie follow these steps:

1. Access the ”Parental Monitoring” feature by navigating to it in the menu tab.
2. Upon selecting the ”Parental Monitoring” option, you will be directed to the Parental Monitoring page. However, to proceed further, you will need to enter the PIN that you created while setting up your account.
3. After successfully entering your PIN, you will be presented with comprehensive analytics detailing the content accessed by your child’s account, including date and time stamps.
4. Additionally, you can review stories generated by AI.
5. This feature is secured by a PIN, ensuring that only parents or authorized individuals can access and monitor the activity.

8.9 Edit Account Details

If you wish to edit your account details, please follow these steps:

1. You can find the ”Edit Account” option by navigating to the Settings icon in the menu tab.
2. Clicking on the ”Edit Account” option will direct you to your profile page. You can also go to the profile page by clicking on the profile icon in the bottom navigation on the home page.
3. On the profile page, you can click on the fields you wish to edit, such as your name, password, or avatar.
4. Edit password will work in the same way as the Forgot Password, you will receive a code in your email, and after the code, you will be prompted to enter the new password and confirm the entered password.
5. Once you have made the desired changes, click on the ”Save” button.
6. Your profile details will be updated with the changes you have made.

8.10 Account Deletion

If you wish to delete your account, please follow these steps:

1. Access the ”Account Deletion” option in the settings menu.
2. You will be prompted to enter your password to confirm the deletion request.
3. After confirming your password, a verification code will be sent to your email address registered with your account.
4. Enter the verification code received in your email to proceed with the account deletion process.
5. Follow the on-screen instructions to confirm the deletion of your account.
6. Once the account deletion process is complete, you will receive a confirmation message, and your account and associated data will be permanently removed from our system.

8.11 Contacting the FictionGenie Team

If you have any questions, feedback, or encounter any issues with the FictionGenie application, you can contact our dedicated support team for assistance. Here’s how:

1. Navigate to the ”Contact Us” section within the application. You can typically find this option in the menu tab or the settings.
2. Click on the ”Contact Us” button to initiate a communication with our support team.
3. Fill out the contact form with your name, email address, and a detailed description of your query or issue.
4. Once you’ve provided all necessary information, submit the form.
5. Our support team will review your inquiry and respond to you as soon as possible, providing assistance or resolving any issues you may have.

We are committed to providing excellent support and ensuring that your experience with FictionGenie is as smooth and enjoyable as possible. Don’t hesitate to reach out to us if you need assistance!

8.12 Conclusion

In this chapter, we have provided a detailed explanation of the user manual for our application FictionGenie. This manual will help users of all ages to get to know about how to use FictionGenie and the

necessary steps. It will help them to get to know the solution of any problem encountered or in explaining steps to their children. We have tried our best to make it as simple as possible so not only parents but children can also comprehend how to use FictionGenie.

Chapter 9 Experimental Results and Discussion

This chapter will discuss the experiments that we have done so far along with the analysis and results of these experiments.

9.1 Pre-built Stories Library

We have experimented with the Pre-built Stories Library module of our system to assess the system performance and the impact on user engagement while using the system. We have added 30 famous children's stories according to different age groups. We have collected these stories according to the 5 main genres mystery, fantasy, fairy tale, science fiction, and historical fiction. The experiment involved the use of an application by 10 users out of which 6 were kids aged between 8 to 10 and 4 were aged between 11 to 15. These users were encouraged to read the stories mainly according to their choice of genres. The results we got after the analysis of user interaction with FicitonGenie's module regarding their interaction, engagement, preferences, and the impact of this module on the overall application are shown below.

9.1.1 User Satisfaction Survey

After the completion of the testing phase when the users have successfully interacted with the application survey was conducted to gather user feedback on the module we have experimented with. These users were asked to rate their satisfaction and overall experience on a scale of 1 to 5.

Average User Satisfaction Rating: 4.6

9.1.2 Popular Genres

Analysis was done to gather insights about those genres that were preferred to read by the majority of the users so we will gather more stories and focus more on this genre in the final deployment.

9.1.3 Feature Popularity

Analysis was done to gather insights about the most used feature within this module other than story reading like searching a story, bookmarking, and genre preferences.

9.1.4 Usage Frequency Analysis

To analyze how often will user visits the application after reading the stories we analyzed how often the user has come back to use the application again after using the Pre-built Stories Library module by the

examination of user logs. This provided us insights regarding the engagement levels and usage patterns among the users.

9.1.5 System Result

In this section, a table is shown to summarize the results of the experiment which involved user interaction with the pre-defined stories module.

Table 9.1: System Results

This table shows the system results that came out after successful experimentation of the Pre-built Stories Library module of the system

Comparison	**Metric**
Total Users	10
Stories Read	30
Average Time Spent per User	20 minutes
Average User Satisfaction Rating	4.6
Most Popular Genre	Mystery (most reads i.e 11)
Most Used Feature	Genre-based Story Search

9.2 AI Generated Stories

We have experimented with the AI-generated stories module of our application and we incorporated the Large Language Model of Google i.e. Gemini. For this experiment, we involved 10 children as this feature was mainly related to them. Users were prompted to enter the title, description, keywords, genre, no of characters, and the difficulty level of the story. Based on their prompts and our prompting engineering mechanisms we were able to generate the stories for users.

9.2.1 Quality of Stories

After the generation of a story, the main challenge was to check the quality of stories generated by AI. The stories generated were read by 10 children and their feedback was taken once they read their own generated stories.

9.2.2 Popular Genres

Experimentation was done to also analyze the genres that children were interested in in the stories and we kept a record of it for the experimentation purpose that we could use further in the development.

9.2.3 User Satisfaction

After the experiment was done and the feedback was taken which involved rating the overall experience on a scale of 1 to 5 the results were very satisfactory as the prompts to the AI model were given with a lot of attention by us making sure that the stories adhere to the ethical and moral values.

Average User Satisfaction Rating: 4.5

Table 9.2: System Results

This table shows the system results that came out after successful experimentation of the AI-Generated Stories module of the system

Comparison	**Metric**
Total Users	10
Stories Generated	30
Stories Generated per User	3
Most Popular Genres	Mystery and Thriller
Average User Satisfaction Rating	4.5

9.3 AI Generated Images

We have experimented with the AI-generated images module of our application and we incorporated the AI-generated images module in the context of story generation. We have integrated the Open AI's model DALLE-3 for image generation based on the content of stories. The images generated are relevant to the scenarios of the story and these images ultimately enhance the reading experience of a child and attract him/her more towards our application. For this experiment, we involved 10 children as this feature was mainly related to them.

9.3.1 Image Relevance

After the generation of a story we carefully analyzed the generated images and their relevance according to the context of the story. Children also analyzed this and their feedback was taken on whether they thought that the images were relevant to the stories.

9.3.2 User Satisfaction

After the experiment was done and the feedback was taken which involved rating the overall experience on a scale of 1 to 5 the results were very much satisfactory as the prompts to the AI model were given with a lot attention by us.

Average User Satisfaction Rating: 4.8

Table 9.3: System Results

This table shows the system results that came out after successful experimentation of the AI-Generated Images module of the system

Comparison	**Metric**
Total Users	10
Stories Generated	30
Images per Story	4
Average User Satisfaction Rating	4.8

9.4 Conclusion

After analysis of our system, we can conclude that till now our system has been successful in producing satisfactory outputs as per the expectations. The results of experimentation of the pre-defined story module and AI-generated stories with images module also showed that users are highly interested in using the complete version of FictionGenie. User feedback will be used to fill those areas where the system is lacking currently and more features will be added as per the user demands.

Chapter 10 Conclusion and Future Work

FictionGenie's team has successfully completed the proposed requirements. The extensive literature review performed, paved a path for selecting the best appropriate Large Language Model for story generation which is Gemini. The dynamic images are generated using DALLE-3 model based on the prompts sent. As this application is focused towards children therefore, we have used multiple filtering techniques in order to ensure age appropriate content to be displayed. The integration of text-to-speech model will allow children to listen the stories if they are unable to read them. As our main target audience are children so we have created an interactive UI along with a scalable architecture to seamlessly add new functionalities to the application as it evolves.

We have empowered our app with Parental Monitoring which includes setting content filters, monitoring usage statistics and receiving insights into their child's interaction with the app. All of these features are necessary as our application is solely aimed towards young children. After our application was developed, we have performed proper testing to ensure that our application is free from bugs and ready to make an impact in the market. All in all, FictionGenie aims to have a huge impact in transforming children's lingual and speaking skills making this application on of a kind.

Moving forward, our future work aims to add multiple new features to the application making it one of a kind. We will be introducing multiple languages in which stories will be generated so that all across the globe users can use them. We are also planning to integrate animated videos generated by AI models according to the story being read. We will be introducing a chat support option so that all user issues are catered and efficiently resolved. Furthermore, story sharing will be added which will allow users to share stories with their friends and peers so that they can also read them. We will make sure our application has zero downtime and will run efficiently. We will show our full dedication and support to this application to make it a globally used app.

Bibliography

[1] A. S. G. Melissa Roemmele and R. Swanson, *LIBSVM: Evaluating Story Generation Systems Using Automated Linguistic Analyses*, 2020. Available: https://roemmele.github.io/publications/fiction_generation.pdf.

[2] M. Cavazza and F. Charles, *LIBSVM:Interactive Storytelling: From AI Experiment To New Media*, 2003. Available: https://www.researchgate.net/profile/Marc-Cavazza/publication/220851669_Interactive_storytelling_from_AI_experiment_to_new_media/links/580f3d0308aef2ef97afbf4b/Interactive-storytelling-from-AI-experiment-to-new-media.pdf.

[3] P. M. Boyd Branch and K. Mathewson, *LIBSVM: Collaborative Storytelling with Human Actors and AI Narrators*, 2021. Available: https://computationalcreativity.net/iccc21/wp-content/uploads/2021/09/ICCC_2021_paper_118.pdf.

[4] A. M. Parag Jain, Priyanka Agrawal and M. Sukhwani, *LIBSVM: Story Generation from Sequence of Independent Short Descriptions*, 2017. Available: https://arxiv.org/abs/1707.05501.

[5] S. L. Amal Alabdulkarim and X. Peng, *LIBSVM:Automatic Story Generation: Challenges and Attempts*, 2021. Available: https://arxiv.org/abs/2102.12634.

[6] N. McIntyre and M. Lapata, *LIBSVM: Learning to Tell Tales: A Data-Driven Approach to Story Generation*, 2009. Available: https://aclanthology.org/P09-1025.pdf.

[7] E. Larios and J. Savage, *LIBSVM: Development of an API to Create Interactive Storytelling Systems*, 2001. Available: https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=a71cb914c0aec908ca53904af21936f62ee31bb8.

[8] ArielHan and Z. Cai, *LIBSVM:Design implications of generative AI systems for visual storytelling for young learners*, 2023. Available: https://dl.acm.org/doi/abs/10.1145/3585088.3593867.

[9] K. E. X. P. Zexin Chen, Eric Zhou and M. Riedl, *LIBSVM: Ambient Adventures: Teaching*

ChatGPT on Developing Complex Stories, 2023. Available: hhttps://arxiv.org/abs/2308.01734.

[10] V. V. N. J. Tianjun Zhang, Yi Zhang and X. Wang, *LIBSVM:Controllable Text to Image Generation with GPT-4*, 2023. Available: https://arxiv.org/abs/2305.18583.

[11] C. App., "Ai story generator and maker." https://play.google.com/store/apps/details?id=com.app.ai.story, 2023. Accessed: October 8, 2023.

[12] S. Spark, "Story spark," 2023. Accessed: 10 8, 2023.

[13] S. Software., "Kidstory: Ai bedtime stories." https://play.google.com/store/apps/details?id=com.shortstori.es, 2023. Accessed: October 7, 2023.

[14] StorySeed.ai, "Story seed." https://play.google.com/store/apps/details?id=com.storyseed.app, 2023. Accessed: October 7, 2023.